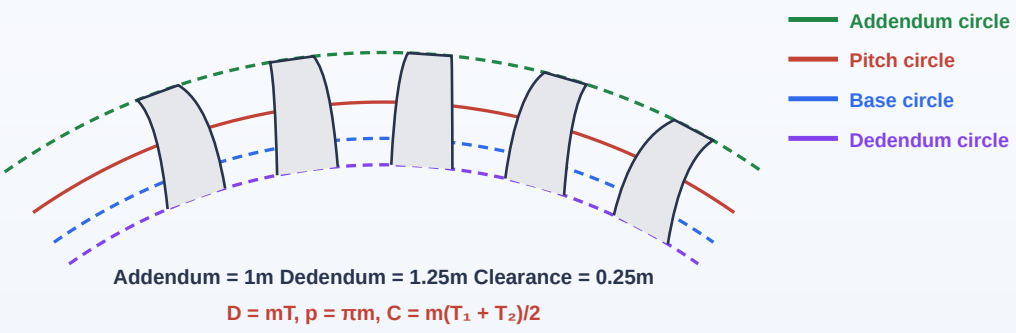
UPSSSC ALERT

UPSSSC JE

MECHANICAL
TOM & MACHINE DESIGN

Complete English Edition

15 Marks · 2026



UPSSSC JE (Mechanical) 2026: Theory of Machines and Machine Design is the most important subject of the paper – **15 questions** out of 65 in Part-1 come from it. This PDF follows every point of the Commission's official syllabus (Advertisement No. 24-Exam/2026).

All topics are compiled in a **simple, systematic and exam-oriented** way. Each chapter contains key points, diagrams, a formula table, detailed explanation, solved numerical problems, 30 important one-liner questions and multiple-choice questions with explanations. Previous year questions of **SSC JE (2018–2023)** and **UPSSSC JE 2016** are also included.

The content is in line with standard BTEUP diploma textbooks (R. S. Khurmi, S. S. Rattan, D. R. Malhotra). A Hindi + English edition of this PDF is also available.

For a better score in JE (Mechanical) 2026, study this PDF and also attempt the QUIZ.

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Chapter 1: Simple Machines

Simple Machines – Basic Concept

A **machine** is a device that enables a large load to be lifted by applying a small effort, or allows a force to be applied at a more convenient point and in a more convenient direction. The lever, pulley blocks, screw jack, wheel and axle, and worm and worm wheel are the common examples of simple machines.

Understand in Detail

1.1 What is a machine — 'more work with less force'

While changing a car wheel we lift a 1000 kg car with a jack using only a small force. We move a heavy stone with a long rod (lever). A device that converts a force (**effort P**) applied at one place into a convenient form for lifting or moving a **load W** at another place is a **simple machine** — such as a lever, pulley, wheel and axle, screw jack, worm and worm wheel.

What is the price paid? A machine does not 'create' energy. If less effort is needed, the effort has to move a **greater distance**. Only after turning the jack handle many times does the car rise a few centimetres. In the ideal case **effort × distance moved by effort = load × distance moved by load**.

Derivation: Relation between MA, VR and efficiency

1. **Mechanical advantage (MA)** = W/P — how many times the load can be lifted (varies due to friction)
2. **Velocity ratio (VR)** = distance moved by effort ÷ distance moved by load = y/x — depends only on geometry, constant
3. Efficiency η = output work/input work = (W × x)/(P × y) = (W/P)/(y/x) → **η = MA/VR**
4. **Ideal machine** (frictionless): η = 1 → MA = VR ; ideal effort P_i = W/VR ; ideal load W_i = P × VR
5. **Effort lost in friction** F_P = P − W/VR ; in terms of load F_W = P × VR − W

Law of machine: if in an experiment the effort needed for different loads is measured and plotted, a nearly straight line is obtained: **P = mW + C**. Here C = effort needed even at zero load (the machine's own friction), m = slope. **Maximum mechanical advantage** = 1/m (C negligible at very large load); **maximum efficiency** = 1/(m × VR).

Reversible vs self-locking machine: if on removing the effort the load comes down by itself (the machine runs backward), it is **reversible**; if the load stays where it is, it is **self-locking (irreversible)**. Condition: **η > 50% → reversible; η < 50% → self-locking**. That is why a screw jack's efficiency is deliberately kept below 50% — so the car does not drop suddenly when the handle is released.

Example: Law of a machine P = 0.05W + 20 (N), VR = 30. At 1000 N load find MA and efficiency, and maximum efficiency.

Solution:

1. P = 0.05 × 1000 + 20 = 70 N ; MA = 1000/70 = **14.29**
2. η = 14.29/30 = **47.6%** (< 50% → self-locking)
3. η_{max} = 1/(mVR) = 1/(0.05 × 30) = **66.7%** → at very large loads the machine becomes reversible

1.2 VR of important simple machines — how it is obtained

Each machine's VR comes from its geometry: move the effort through one full cycle and see how far the load moves.

Derivation: Standard VR formulae

1. **Lever:** VR = effort arm/load arm ; three classes — first (fulcrum in the middle: scissors, see-saw), second (load in the middle: wheelbarrow, nutcracker), third (effort in the middle: tongs, human forearm — MA < 1)
2. **Wheel and axle:** effort on wheel (D), load on axle rope (d) → VR = D/d ; **differential wheel and axle:** VR = $2D/(d_1 - d_2)$
3. **Weston differential pulley:** VR = 2D/(D − d) — the smaller the difference between the two pulleys, the larger the VR
4. **Pulley systems:** first (n movable pulleys) VR = 2^n ; second (n pulleys in all) VR = n ; third VR = $2^n - 1$
5. **Screw jack:** in one turn of the handle (radius L) the load rises one pitch p → **VR = 2πL/p** ; **differential screw jack:** VR = $2\pi L/(p_1 - p_2)$
6. **Worm and worm wheel:** VR = (effort wheel radius × teeth on worm wheel T)/(load drum radius × starts on worm n) — very high VR, usually self-locking

Example: A screw jack has handle length 40 cm and screw pitch 1 cm. VR? If 100 N effort lifts 5000 N, efficiency?

Solution:

1. VR = 2π × 40/1 = **251.3**
2. MA = 5000/100 = 50 → η = 50/251.3 = **19.9%** — self-locking, as a jack should be

1. Key Points

- **Mechanical Advantage (MA):** the ratio of the load lifted (W) to the effort applied (P), i.e. MA = W/P. It is a dimensionless quantity.
- **Velocity Ratio (VR):** the ratio of the distance moved by the effort (y) to the distance moved by the load (x), i.e. VR = y/x. VR depends only on the geometry of the machine, not on friction.

- **Efficiency (η):** the ratio of output work to input work, i.e. η = MA/VR. In an **ideal machine** friction is zero, so η = 100% and MA = VR.
- **Ideal effort and ideal load:** P_0 = W/VR and W_0 = P × VR. Friction in terms of effort = P − P_0; friction in terms of load = W_0 − W.
- **Law of machine:** P = mW + C, where m is the slope of the P–W line and C is the effort needed to overcome friction (effort at zero load).
- **Maximum MA and maximum efficiency:** MA_{max} = 1/m and η_{max} = 1/(m × VR).
- **Reversible and self-locking machines:** a machine with efficiency above 50% is **reversible** – the load comes down by itself when the effort is removed. A machine with efficiency below 50% is **self-locking** (irreversible), e.g. screw jack, worm and worm wheel.
- **Classes of lever:** First class – fulcrum in the middle (scissors, see-saw). Second class – load in the middle (nut-cracker, wheelbarrow), MA > 1. Third class – effort in the middle (tongs, human forearm), MA < 1.

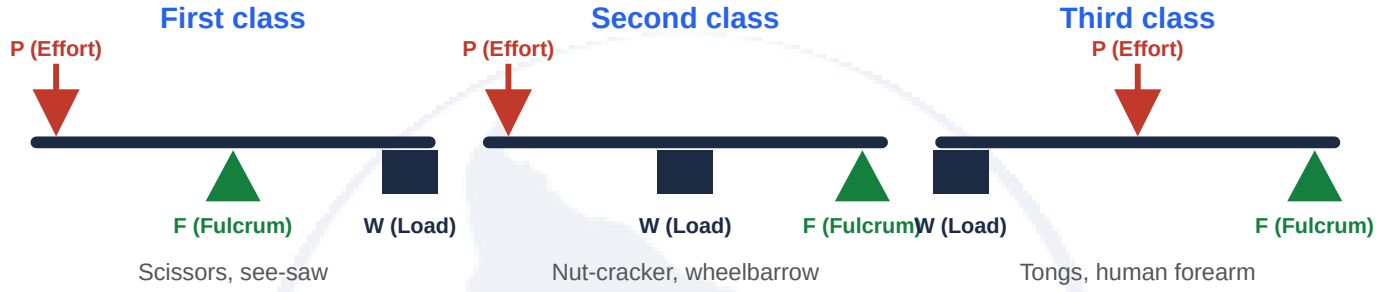


Fig 1.1: Three classes of lever

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Simple wheel and axle	**VR = D / d**
Differential wheel and axle	**VR = 2D / (d_1 − d_2)**
Weston differential pulley block	**VR = 2D / (D − d)**
First system of pulleys (n pulleys)	**VR = 2^n**
Second system of pulleys	**VR = n**
Third system of pulleys	**VR = 2^n − 1**
Simple screw jack	**VR = 2πL / p**
Worm and worm wheel (single start)	**VR = L·T / r**
Efficiency of screw jack	**η = tan α / tan(α + φ)**
Maximum efficiency of screw jack	**η_{max} = (1 − sin φ) / (1 + sin φ)**

3. Detailed Explanation

- **Ideal vs actual machine:** an ideal machine has no friction, so output work equals input work. In an actual machine part of the effort is spent in overcoming friction; hence MA is less than VR and efficiency is less than 1.
- **Why VR is independent of friction:** VR is only a ratio of distances, fixed by geometric quantities such as pulley diameter, screw pitch or lever length. Friction changes the forces, not the distances moved.
- **Graph of the law of machine:** the P–W graph is a straight line. As load increases, MA = W/(mW + C) increases and approaches 1/m. Therefore a machine is more efficient at higher loads.
- **Analysis of a screw jack:** a screw is an inclined plane wrapped round a cylinder, with tan α = p/(πd). Force at the screw to raise the load P = W·tan(α + φ); ideal force P_0 = W·tan α. Effort at the end of the lever = P × (d/2)/L.
- **Condition for self-locking:** force required to lower the load = W·tan(φ − α). If φ > α this force is positive, i.e. the load will not come down by itself. This is the self-locking condition, and the efficiency is then below 50%.
- **Worm and worm wheel:** for one revolution of an n-start worm, a worm wheel with T teeth turns n/T revolution. Hence VR = L·T/(r·n), where L is the radius of the effort wheel and r is the radius of the load drum.

4. Comparison Table — Reversible vs Self-locking Machine

Basis	Reversible	Self-locking
Efficiency	More than 50%	Less than 50%
When effort is removed	Load comes down by itself	Load stays in position
Screw condition	φ < α	φ > α
Examples	Pulley block, wheel and axle	Screw jack, worm and worm wheel

5. Solved Numerical Problems

Example 1. A machine lifts a load of 1000 N with an effort of 60 N and a load of 2000 N with an effort of 100 N. Its velocity ratio is 40. Find the law of the machine, the MA and efficiency at 3000 N, and the maximum efficiency.

Solution:

1. m = (100 − 60)/(2000 − 1000) = 0.04; C = 60 − 0.04 × 1000 = 20 N, hence **P = 0.04W + 20**
2. At W = 3000 N, P = 140 N, so MA = 3000/140 = **21.43** and η = 21.43/40 = **53.6%**
3. MA_{max} = 1/m = 25; η_{max} = 1/(m × VR) = 1/(0.04 × 40) = **62.5%** (the machine is reversible)

Example 2. A screw jack has pitch 10 mm, mean diameter 50 mm and coefficient of friction 0.1. Find the force required at the screw to raise a load of 10 kN and the efficiency. Is the jack self-locking?

Solution:

1. $\tan \alpha = p/(\pi d) = 10/(\pi \times 50) = 0.0637$, so $\alpha = 3.64°$; $\tan \varphi = 0.1$, so $\varphi = 5.71°$
2. $P = W \tan(\alpha + \varphi) = 10000 \times \tan 9.35° = 10000 \times 0.1646 =$ **1646 N**
3. $\eta = \tan \alpha / \tan(\alpha + \varphi) = 0.0637/0.1646 =$ **38.7%**
4. Since $\varphi > \alpha$ and $\eta < 50\%$, the jack is **self-locking**

Example 3. In a differential wheel and axle, the wheel diameter is 500 mm and the two parts of the axle are 200 mm and 180 mm in diameter. Find the effort required to lift 5000 N at 60% efficiency.

Solution:

1. $VR = 2D/(d_1 - d_2) = 1000/20 =$ **50**
2. $MA = \eta \times VR = 0.6 \times 50 = 30$
3. $P = W/MA = 5000/30 =$ **166.7 N**

6. Frequently Asked Facts

- Definitions: MA = W/P, VR = y/x and η = MA/VR.
- When friction increases, VR remains unchanged, but MA and efficiency decrease.
- A screw jack is self-locking only when the friction angle φ is greater than the helix angle α.

7. Note (Confusing Facts)

- In an actual machine MA is always less than VR, so efficiency is less than 1.
- A third-class lever has MA < 1, yet it is useful because it gives a gain in speed (movement).
- The denominator of the differential wheel-and-axle formula has the **difference** of diameters ($d_1 - d_2$), not the sum.

Remember: One line: a self-locking machine always has efficiency below 50%. Assertion–Reason: (A) A screw jack does not come down by itself when the effort is removed. (R) Its efficiency is less than 50%. — Both A and R are true and R is the correct explanation of A.

30 Most Important One-Liner Facts — Simple Machines

1. Formula for efficiency of a machine?

— MA ÷ VR (η = MA/VR).

2. Relation between MA and VR in an ideal machine?

— Both are equal (MA = VR).

3. Unit of mechanical advantage?

— No unit – it is a ratio.

4. On what does velocity ratio depend?

— Only on the geometry of the machine.

5. Effect of friction on VR?

— None – VR stays the same.

6. Law of machine?

— P = mW + C.

7. What does C represent in the law of machine?

— Effort lost in friction (effort at zero load).

8. Maximum mechanical advantage?

— $MA_{max} = 1/m$.

9. Maximum efficiency of a machine?

— $\eta_{max} = 1/(m \times VR)$.

10. Condition for a reversible machine?

— Efficiency more than 50%.

11. Condition for a self-locking machine?

— Efficiency less than 50%.

12. Two examples of self-locking machines?

— Screw jack and worm–worm wheel.

13. Ideal effort?

— $P_0 = W/VR$.

14. Friction in terms of effort?

— $P - P_0$.

15. Lever with fulcrum in the middle?

— First-class lever.

16. Lever with load in the middle?

— Second-class lever (MA > 1).

17. Lever with effort in the middle?

— Third-class lever (MA < 1).

18. Class of lever of a nut-cracker?

— Second class.

19. Class of lever of the human forearm?

— Third class.

20. VR of a simple wheel and axle?

— D/d.

21. VR of a differential wheel and axle?

— $2D/(d_1 - d_2)$.

22. VR of a Weston differential pulley block?

— $2D/(D - d)$.

23. VR of the first system of pulleys?

— 2^n.

24. VR of the second system of pulleys?

— n.

25. VR of the third system of pulleys?

— $2^n - 1$.

26. VR of a simple screw jack?

— $2\pi L/p$.

27. Efficiency of a screw jack?

— $\tan \alpha / \tan(\alpha + \varphi)$.

28. Maximum efficiency of a screw jack?

— $(1 - \sin \varphi)/(1 + \sin \varphi)$.

29. Self-locking condition of a screw?

— Friction angle φ > helix angle α.

30. Effect of increasing load on efficiency?

— Efficiency increases and approaches η_{max}.

Practice MCQs — Simple Machines

Q1. The efficiency of an ideal machine is:

(a) 50%

(b) 75%

(c) 100%

(d) More than 100%

✔ Answer: (c) 100%

Explanation: In an ideal machine friction is zero, so output work = input work and $\eta = 100\%$.

Q2. In the law of machine P = mW + C, C represents:

(a) Load
(b) Effort to overcome friction
(c) Velocity ratio
(d) Efficiency

✔ **Answer: (b) Effort to overcome friction**

Explanation: At W = 0 an effort C is still needed; it is spent only in overcoming friction.

Q3. The efficiency of a self-locking machine is:

(a) Less than 50%
(b) Exactly 50%
(c) More than 50%
(d) 100%

✔ **Answer: (a) Less than 50%**

Explanation: Self-locking means the load does not run back; this happens only when $\eta < 50\%$.

Q4. The velocity ratio of a simple screw jack is:

(a) $p/2\pi L$
(b) $2\pi L/p$
(c) $\pi L/p$
(d) L/p

✔ **Answer: (b) $2\pi L/p$**

Explanation: For one turn of the lever the effort moves $2\pi L$ and the load rises by one pitch p, so $VR = 2\pi L/p$.

Q5. In the third system of pulleys with 4 pulleys, the velocity ratio is:

(a) 8
(b) 15
(c) 16
(d) 4

✔ **Answer: (b) 15**

Explanation: Third system: $VR = 2^n - 1 = 2^4 - 1 = 15$.

Q6. A nut-cracker is a lever of which class?

(a) First
(b) Second
(c) Third
(d) None of these

✔ **Answer: (b) Second**

Explanation: In a nut-cracker the load (nut) lies between the fulcrum and the effort – second-class lever.

Q7. Maximum efficiency of a screw jack is:

(a) $(1 - \sin\varphi)/(1 + \sin\varphi)$
(b) $(1 + \sin\varphi)/(1 - \sin\varphi)$
(c) $\tan\varphi$
(d) $\cos\varphi$

✔ **Answer: (a) $(1 - \sin\varphi)/(1 + \sin\varphi)$**

Explanation: Setting the derivative of $\eta = \tan\alpha/\tan(\alpha + \varphi)$ to zero gives $\eta_{max} = (1 - \sin\varphi)/(1 + \sin\varphi)$.

Q8. Which quantity remains unchanged when friction increases?

(a) Mechanical advantage
(b) Efficiency
(c) Velocity ratio
(d) Effort

✔ **Answer: (c) Velocity ratio**

Explanation: VR is a ratio of distances fixed by geometry; friction only changes forces.

Q9. The velocity ratio of a Weston differential pulley block is:

(a) 2D/(D − d)
(b) D/(D − d)
(c) 2D/(D + d)
(d) (D − d)/2D

✔ **Answer: (a) 2D/(D − d)**

Explanation: For a Weston differential pulley block, VR = 2D/(D − d).

Q10. If the law of machine is P = 0.05W + 10, the maximum mechanical advantage is:

(a) 10
(b) 20
(c) 50
(d) 5

✔ **Answer: (b) 20**

Explanation: $MA_{max} = 1/m = 1/0.05 = 20$.

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Chapter 2: Four-bar Linkage & Link Motion

Four-bar Linkage & Link Motion – Basic Concept

A **link** (kinematic link) is a resistant body that forms part of a machine and has relative motion with respect to other parts. Two links in contact form a **kinematic pair**; links joined so that relative motion is possible form a **kinematic chain**. When one link of a chain is fixed, it becomes a **mechanism**. The four-bar chain is the simplest and most important of all chains.

Understand in Detail

2.1 Link, pair, chain and mechanism — the 'bones' of a machine

Every machine is made of many parts that move relative to each other in a definite way. **Link (element)** — a resistant part of a machine that moves relative to another part (connecting rod, crank, piston); a link may be rigid, flexible (belt, rope — tension only) or fluid (oil in a hydraulic press — compression only). **Kinematic pair** — a joint of two links such that their relative motion is constrained.

Classification of pairs: by type of motion — sliding (piston-cylinder), turning (shaft-bearing), rolling (ball bearing), screw (nut-bolt), spherical (ball and socket). **By type of contact** — **lower pair**: surface contact (sliding, turning, screw); **higher pair**: point or line contact (cam-follower, gear teeth, ball bearing). **By closure** — self-closed and force-closed (cam-follower held by a spring).

Kinematic chain: a closed group of links in which every motion is definite. When one link of a chain is **fixed** it becomes a **mechanism**; and when a mechanism does useful work, a **machine**. **Inversion** — obtaining different mechanisms by fixing different links of the same chain; a chain of n links has n inversions.

Derivation: Degree of freedom — Grubler's equation

1. A free rigid body in a plane has 3 degrees of freedom (x, y, rotation)
2. n links with one fixed → total freedom of moving links = $3(n - 1)$
3. Each lower pair (j) removes 2 freedoms, each higher pair (h) removes 1
4. **$F = 3(n - 1) - 2j - h$** (Kutzbach) ; $F = 1$ → mechanism (constrained motion) ; $F = 0$ → structure (no motion) ; $F < 0$ → over-constrained structure ; $F > 1$ → unconstrained motion
5. Example: four-bar $n = 4$, $j = 4$ → $F = 9 - 8 =$ **1** ✔ ; three links $n = 3$, $j = 3$ → $F = 6 - 6 = 0$ (triangle — a structure)

2.2 Four-bar chain, Grashof's law and inversions

The **four-bar chain** is the simplest and most useful mechanism: four rigid links, four turning pairs. Names of links — **fixed link (frame)**, **crank** (rotates a full 360°), **coupler** (the middle one) and **lever/rocker** (only oscillates).

Grashof's law: if the sum of the shortest (s) and longest (l) links is less than or equal to the sum of the other two (p, q) — **s + l ≤ p + q** — at least one link can make a full revolution (Grashof chain). Which link is fixed then decides the mechanism: **a link adjacent to the shortest fixed → crank-rocker** (the shortest becomes the crank); **the shortest itself fixed → double-crank (drag link)**; **the link opposite the shortest fixed → double-rocker**. If s + l > p + q, no link rotates fully in any inversion — always double-rocker.

Inversions of the four-bar (examples): beam engine (crank-rocker), coupling rod of locomotive wheels (double-crank, parallelogram), Watt's approximate straight-line mechanism and Ackermann steering (double-rocker).

> **Example:** Links are 2, 5, 4, 6 cm and the 4 cm link is fixed. What mechanism?
> **Solution:**
> 1. s = 2, l = 6 → s + l = 8 ; p + q = 5 + 4 = 9 → 8 < 9 → **Grashof chain**
> 2. If the 4 cm link is adjacent to the shortest (2) → **crank-rocker** (2 cm crank) ; if opposite → double-rocker

2.3 Single and double slider-crank chains — from engines to shapers

Replacing one turning pair of a four-bar chain with a **sliding pair** gives the **single slider-crank chain** (3 turning + 1 sliding). Its four inversions: (1) cylinder (frame) fixed → **reciprocating engine/compressor**; (2) crank fixed → **Whitworth quick-return** mechanism, **rotary (Gnome) engine**; (3) connecting rod fixed → **oscillating cylinder engine**, **crank and slotted lever quick-return** (shaper machine); (4) slider (piston) fixed → **hand pump** (pendulum pump).

Meaning of quick return: in a shaper the tool cuts metal on the forward stroke (slowly, more force) and cuts nothing on return — so the return stroke is made faster to save time. **Time ratio** = time of cutting stroke ÷ time of return = $(360° - \beta)/\beta$, where β = angle turned by the crank during return.

Double slider-crank chain (2 turning + 2 sliding) inversions: **elliptical trammel** — drawing ellipses; **Scotch yoke** — converting rotary into simple harmonic motion; **Oldham coupling** — joining two parallel but slightly offset shafts.

> **Note: Instantaneous centre:** the point about which a moving link appears to be in pure rotation at that instant. In a mechanism of n links the number of instantaneous centres = **n(n − 1)/2** ; 6 in a four-bar. **Kennedy's theorem:** the three instantaneous centres of three links moving relative to one another lie on a straight line.

1. Key Points

- **Kinematic pair:** two links in contact having relative motion. **Lower pair** – surface contact (sliding, turning, screw pair). **Higher pair** – line or point contact (cam–follower, gears, ball bearing).
- **Closed pairs:** self-closed (held by geometry, e.g. pin joint) and force-closed (held by an external force, e.g. cam and follower held by a spring).

- **Kutzbach equation (planar):** $n = 3(l - 1) - 2j - h$, where l = links, j = binary joints (lower pairs), h = higher pairs. A ternary joint counts as 2 binary joints.
- **Grübler's criterion:** for a constrained mechanism with only lower pairs, $3l - 2j - 4 = 0$.
- **Degree of freedom:** n = 0 – structure; n = 1 – constrained mechanism; n > 1 – unconstrained; n < 0 – super-structure (statically indeterminate).
- **Grashof's law:** if $s + l \leq p + q$ (s = shortest, l = longest link), at least one link can make a full revolution.
- **Inversions of a Grashof chain:** shortest link fixed – double crank; link adjacent to shortest fixed – crank–rocker; link opposite to shortest fixed – double rocker.
- **Inversion:** a different mechanism obtained by fixing different links of the same kinematic chain. A four-bar chain and a single slider crank chain each have 4 inversions.
- **Instantaneous centre:** the point about which a link has pure rotation at a given instant. Number of I-centres $N = n(n - 1)/2$. Kennedy's theorem: three bodies in plane motion have three I-centres lying on one straight line.
- **Coriolis acceleration** ($2\omega v$) appears when a point slides on a link that is rotating; its direction is obtained by rotating the sliding velocity through 90° in the sense of ω.

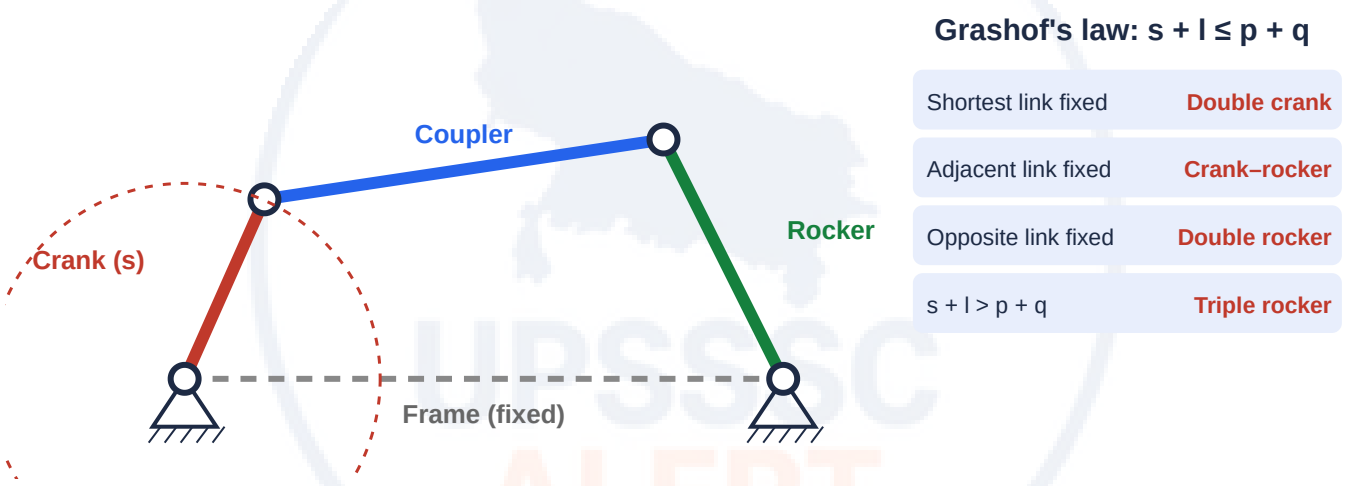


Fig 2.1: Four-bar linkage and inversions by Grashof's law

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Single slider crank – 1st inversion	**Reciprocating engine / compressor**
2nd inversion (crank fixed)	**Whitworth quick-return, rotary (Gnome) engine**
3rd inversion (connecting rod fixed)	**Crank and slotted lever (shaper), oscillating cylinder engine**
4th inversion (slider fixed)	**Hand pump, pendulum pump (bull engine)**
Double slider crank chain	**Elliptical trammel, Scotch yoke, Oldham's coupling**
Four-bar inversions	**Beam engine, locomotive coupling rod, Watt's indicator**
Number of instantaneous centres	**$N = n(n - 1)/2$**
Coriolis acceleration	**$a_c = 2\omega v$**

3. Detailed Explanation

- **Kinematic chain condition:** for a chain with only lower pairs, $l = 2p - 4$ and $j = (3/2)l - 2$, where p is the number of pairs. If the left side exceeds the right it is a locked chain; if it is less, the chain is unconstrained.
- **Slider crank chain:** it is a four-bar chain in which one turning pair is replaced by a sliding pair. Fixing the cylinder gives the I.C. engine; fixing the crank gives the Whitworth quick-return; fixing the connecting rod gives the crank and slotted lever; fixing the slider gives the hand pump.
- **Quick-return mechanism:** in a shaper the cutting stroke takes more time than the return stroke. Time ratio = β/α = cutting angle/return angle, where $\cos(\alpha/2) = r/c$ (crank radius/distance between fixed centres).
- **Straight-line mechanisms:** exact – Peaucellier, Hart, Scott-Russell; approximate – Watt, Grasshopper, Tchebicheff, Robert.
- **Pantograph** reproduces a path to an enlarged or reduced scale; **Ackermann** and **Davis** steering gears give correct steering, the condition being $\cot \varphi - \cot \theta = c/b$.
- **Hooke's joint (universal coupling)** connects two intersecting shafts; the velocity ratio fluctuates within each revolution, and a double Hooke's joint gives a constant velocity ratio.

4. Comparison Table — Lower Pair vs Higher Pair

Basis	Lower Pair	Higher Pair
Type of contact	Surface contact	Line or point contact
Wear	Less	More
Examples	Piston–cylinder, nut–bolt, hinge	Cam–follower, gears, ball bearing
Effect in Kutzbach equation	Removes 2 DOF (−2j)	Removes 1 DOF (−h)

5. Solved Numerical Problems

Example 1. A planar mechanism has 6 links and 7 lower pairs (binary joints) and no higher pair. Find its degree of freedom.

Solution:

1. $n = 3(l - 1) - 2j - h = 3(6 - 1) - 2 \times 7 - 0 = 15 - 14 =$ **1**
2. The degree of freedom is 1, so it is a **constrained mechanism**

Example 2. Find the number of instantaneous centres in a mechanism with 6 links.

Solution:

1. $N = n(n - 1)/2 = 6 \times 5/2 =$ **15**

Example 3. The links of a four-bar chain, in order, are 50, 80, 100 and 120 mm. Which mechanism is obtained when the 80 mm link is fixed?

Solution:

1. $s + l = 50 + 120 = 170$; $p + q = 80 + 100 = 180$, so $s + l < p + q$ – it is a **Grashof chain**
2. The 80 mm link is adjacent to the shortest (50 mm) link, so a **crank–rocker mechanism** is obtained

Example 4. In a crank and slotted lever mechanism the crank radius is 150 mm and the distance between fixed centres is 300 mm. Find the quick-return ratio.

Solution:

1. $\cos(\alpha/2) = r/c = 150/300 = 0.5$, so $\alpha/2 = 60°$ and $\alpha = 120°$ (return stroke)
2. Cutting stroke $\beta = 360° - 120° = 240°$
3. Ratio $= \beta/\alpha = 240/120 =$ **2 : 1**

6. Frequently Asked Facts

- Kutzbach: $n = 3(l - 1) - 2j - h$; DOF = 1 means a constrained mechanism.
- Number of instantaneous centres $N = n(n - 1)/2$; Kennedy's theorem – three I-centres lie on a straight line.
- Oldham's coupling, Scotch yoke and elliptical trammel are inversions of the double slider crank chain.
- Coriolis acceleration = $2\omega v$, appears only when there is sliding on a rotating link.

7. Note (Confusing Facts)

- Elliptical trammel is NOT an inversion of the single slider crank chain – it belongs to the double slider crank chain.
- A ternary joint is counted as **two** binary joints in Kutzbach's equation.
- Fixing the shortest link of a Grashof chain gives a double crank, not a crank–rocker.

Remember: One line: a structure has zero degree of freedom; a constrained mechanism has one. Assertion–Reason: (A) Cam and follower form a higher pair. (R) They have point or line contact. — Both true; R explains A.

30 Most Important One-Liner Facts — Four-bar Linkage & Link Motion

1. A body having relative motion in a machine is called?

— Kinematic link (element).

2. Two links in contact with relative motion form?

— A kinematic pair.

3. Type of contact in a lower pair?

— Surface contact.

4. Type of contact in a higher pair?

— Line or point contact.

5. Example of a higher pair?

— Cam–follower, gears, ball bearing.

6. Example of a sliding pair?

— Piston and cylinder.

7. A nut and bolt form which pair?

— Screw (helical) pair.

8. Kinematic chain with one link fixed is called?

— Mechanism.

9. Kutzbach equation for plane mechanisms?

— $n = 3(l - 1) - 2j - h$.

10. Grübler's criterion?

— $3l - 2j - 4 = 0$.

11. DOF of a structure?

— Zero.

12. DOF of a constrained mechanism?

— One.

13. DOF of a free rigid body in space?

— Six.

14. Grashof's law?

— $s + l \le p + q$.

15. Shortest link fixed in a Grashof chain gives?

— Double crank mechanism.

16. Link adjacent to the shortest fixed gives?

— Crank–rocker mechanism.

17. Link opposite to the shortest fixed gives?

— Double rocker mechanism.

18. Number of inversions of a four-bar chain?

— Four.

19. Inversion used in a reciprocating engine?

— First inversion of slider crank chain.

20. Whitworth quick-return is which inversion?

— Second (crank fixed).

21. Crank and slotted lever is which inversion?

— Third (connecting rod fixed).

22. Hand pump is which inversion?

— Fourth (slider fixed).

23. Oldham's coupling is an inversion of?

— Double slider crank chain.

24. Example of an exact straight-line mechanism?

— Peaucellier mechanism.

25. Example of an approximate straight-line mechanism?

— Watt's mechanism.

26. Number of instantaneous centres?

— $N = n(n - 1)/2$.

27. Kennedy's theorem?

— Three I-centres of three bodies lie on one straight line.

28. Magnitude of Coriolis acceleration?

— $2\omega v$.

29. Quick-return ratio of a shaper?

— Cutting time ÷ return time = β/α.

30. Hooke's joint connects which shafts?

— Two intersecting shafts.

Practice MCQs — Four-bar Linkage & Link Motion

Q1. Cam and follower form which type of pair?

(a) Lower, self-closed

(b) Higher, force-closed

(c) Lower, force-closed

(d) Turning pair

✔ **Answer: (b) Higher, force-closed**

Explanation: Cam and follower have point/line contact (higher pair) and are kept in contact by a spring – force-closed.

Q2. An assembly with zero degree of freedom is called:

(a) Mechanism

(b) Structure

(c) Machine

(d) Chain

✔ **Answer: (b) Structure**

Explanation: With DOF = 0 no relative motion is possible – the assembly is a structure.

Q3. How many inversions does a four-bar chain have?

(a) 2

(b) 3

(c) 4

(d) 6

✔ **Answer: (c) 4**

Explanation: A four-bar chain has four links, so four inversions are possible by fixing each link in turn.

Q4. In a Grashof chain, fixing the shortest link gives:

(a) Double crank

(b) Crank-rocker

(c) Double rocker

(d) Triple rocker

✔ **Answer: (a) Double crank**

Explanation: Fixing the shortest link lets both adjacent links rotate fully – double crank.

Q5. Oldham's coupling is an inversion of:

(a) Four-bar chain

(b) Single slider crank

(c) Double slider crank

(d) Kinematic pair

✔ **Answer: (c) Double slider crank**

Explanation: Oldham's coupling is obtained by fixing one of the links of the double slider crank chain.

Q6. A hand pump is based on which inversion?

(a) Slider fixed

(b) Crank fixed

(c) Connecting rod fixed

(d) Cylinder fixed

✔ **Answer: (a) Slider fixed**

Explanation: In a hand pump the slider (pump barrel) is fixed – fourth inversion.

Q7. Which is an exact straight-line mechanism?

(a) Watt
(b) Peaucellier
(c) Tchebicheff
(d) Robert

✔ **Answer: (b) Peaucellier**

Explanation: The Peaucellier mechanism generates an exact straight line; Watt's gives an approximate one.

Q8. Number of instantaneous centres in an 8-link mechanism:

(a) 16
(b) 24
(c) 28
(d) 32

✔ **Answer: (c) 28**

Explanation: $N = n(n - 1)/2 = 8 \times 7/2 = 28$.

Q9. The magnitude of Coriolis acceleration is:

(a) ωv
(b) $2\omega v$
(c) $\omega^2 r$
(d) v^2/r

✔ **Answer: (b) $2\omega v$**

Explanation: Coriolis acceleration = $2\omega v$, perpendicular to the sliding velocity.

Q10. In Kutzbach's equation, a ternary joint is counted as:

(a) 1 binary joint
(b) 2 binary joints
(c) 3 binary joints
(d) 0

✔ **Answer: (b) 2 binary joints**

Explanation: A ternary joint joins three links, which is equivalent to two binary joints.

SSC JE (Mechanical) Previous Year Questions — Four-bar Linkage & Link Motion

Q1. Relation between number of joints (j) and links (l) in a kinematic chain:

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(a) $j = (3/2)l - 2$
(b) $j = (2/3)l + 2$
(c) $j = 2l - 4$
(d) $j = (3/2)l + 2$

✔ **Answer: (a) $j = (3/2)l - 2$**

Explanation: For a chain with only lower pairs, $j = (3/2)l - 2$.

Q2. Scotch yoke mechanism is an inversion of:

SSC JE 2018 (23 January, Evening)

(a) Four-bar chain
(b) Single slider crank chain
(c) Double slider crank chain
(d) Crossed slider crank chain

✔ **Answer: (c) Double slider crank chain**

Explanation: Scotch yoke is obtained by fixing a slider of the double slider crank chain.

Q3. Which is NOT an inversion of a single slider crank chain?

SSC JE 2018 (23 January, Evening)

(a) Whitworth quick return
(b) Oscillating cylinder engine
(c) Elliptical trammel
(d) Hand pump

✔ **Answer: (c) Elliptical trammel**

Explanation: The elliptical trammel belongs to the double slider crank chain.

Q4. Type of contact in a higher pair:

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(a) Surface contact
(b) Point (or line) contact
(c) Volume contact
(d) No contact

✔ **Answer: (b) Point (or line) contact**

Explanation: Higher pairs have point or line contact.

Q5. Degrees of freedom of a free body in space:

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(a) 3
(b) 4
(c) 5
(d) 6

✔ **Answer: (d) 6**

Explanation: A free body in space has 3 translations and 3 rotations – 6 DOF.

Q6. A linkage becomes a structure when its DOF is:

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(a) 0
(b) 1
(c) 2
(d) 3

✔ **Answer: (a) 0**

Explanation: With zero DOF no motion is possible – a structure.

Q7. Inversion of a mechanism is obtained by:

SSC JE 2018 (23 January, Morning)

(a) Changing higher pair into lower pair
(b) Fixing different links of a kinematic chain
(c) Turning it upside down
(d) Reversing input and output motion

✔ **Answer: (b) Fixing different links of a kinematic chain**

Explanation: Inversions are obtained by fixing, one at a time, different links of a chain.

Q8. Oldham coupling is an inversion of:

SSC JE (Previous Year)

(a) Four-bar linkage
(b) Six-bar linkage
(c) Single slider crank
(d) Double slider crank

✔ **Answer: (d) Double slider crank**

Explanation: Oldham's coupling – double slider crank chain inversion.

Q9. A universal joint is an example of:

SSC JE (Previous Year)

(a) Lower pair
(b) Higher pair
(c) Rolling pair
(d) Sliding pair

✔ **Answer: (a) Lower pair**

Explanation: The pins and cross of a universal joint form turning pairs, which are lower pairs.

Q10. If crank radius of a slider crank increases from 10 cm to 12 cm, stroke length increases by: SSC JE (Previous Year)

(a) 20%
(b) 12%
(c) 15%
(d) 10%

✔ **Answer: (a) 20%**

Explanation: Stroke = 2r; it rises from 20 cm to 24 cm, i.e. an increase of 20%.

Q11. Coriolis acceleration (cm/s²) of a slider moving at 10 cm/s on a link rotating at 60 rpm: SSC JE (Previous Year)

(a) 62.8
(b) 125.6
(c) 31.4
(d) 251.2

✔ **Answer: (b) 125.6**

Explanation: $\omega = 2\pi \times 60/60 = 6.28$ rad/s; $a_c = 2\omega v = 2 \times 6.28 \times 10 = 125.6$ cm/s².

UPSSSC ALERT

Chapter 3: Flywheel & Fluctuation of Energy

Flywheel & Fluctuation of Energy – Basic Concept

A **flywheel** is a heavy rotating wheel mounted on the crankshaft that acts as a **reservoir of energy**. When the energy supplied exceeds the requirement it stores the excess and its speed rises; when the supply is less it gives back energy and its speed falls. It thus keeps the fluctuation of speed within a cycle within desired limits.

Understand in Detail

3.1 Why a flywheel — the engine's 'uneven breathing'

In a four-stroke engine power is obtained in only one of the four strokes; in the other three the engine actually spends energy (compression, exhaust, suction). So the engine's **torque keeps varying**, while the machine driven (e.g. a generator) needs nearly constant torque. If nothing is done, the shaft will speed up during the power stroke and slow down in the rest — jerky motion.

A **flywheel** is a heavy wheel mounted on the crankshaft that acts as a **reservoir of energy**: when the engine gives more energy than required, the flywheel speeds up slightly and stores it as kinetic energy; when it gives less, the flywheel slows down slightly and returns that energy. In this way the speed fluctuation **within a cycle** stays limited.

Flywheel vs governor — a favourite exam question: a flywheel controls the speed fluctuation **within a cycle**, not the mean speed when load changes. A **governor** controls the **mean speed** over many cycles by changing the fuel/steam supply when load changes. A flywheel has nothing to do with fuel supply; a governor directly does.

3.2 Turning moment diagram and fluctuation of energy

Turning moment (crank-effort) diagram — a graph of torque on the crankshaft (vertical) against crank angle (horizontal). The area under it = work (energy). Draw a horizontal line for the **mean resisting torque** — where the diagram is **above** this line there is excess energy (the flywheel speeds up), where **below** there is a deficit (the flywheel slows down).

Maximum fluctuation of energy ΔE = difference of flywheel energy at the points of maximum and minimum speed = the difference between the maximum and minimum cumulative values when the successive areas (+/−) are added.

Derivation: $\Delta E = I\,\omega^2\,C_s$

1. Kinetic energy of flywheel = ½ I ω² (I = moment of inertia = m k²)
2. $\Delta E = \frac{1}{2} I (\omega_{max}^2 - \omega_{min}^2) = \frac{1}{2} I (\omega_{max} + \omega_{min})(\omega_{max} - \omega_{min})$
3. Mean $\omega = (\omega_{max} + \omega_{min})/2 \rightarrow \Delta E = I\,\omega\,(\omega_{max} - \omega_{min})$
4. **Coefficient of fluctuation of speed $C_s = (\omega_{max} - \omega_{min})/\omega \rightarrow \Delta E = I\,\omega^2\,C_s = m\,k^2\,\omega^2\,C_s = 2E\,C_s$**
5. **Meaning:** to keep C_s small (≈ 0.002–0.01 in generators) a large I is needed ; increasing ω handles more ΔE for the same I — so high-speed engines have lighter flywheels

Coefficient of fluctuation of energy C_E = ΔE ÷ work done per cycle. It is highest in a single-cylinder four-stroke and much lower in a multi-cylinder engine — so a multi-cylinder engine needs a smaller flywheel. A **rim-type flywheel** is most effective since the mass is outside (large k); I = m R² (approx.). Hoop stress in the rim σ = ρ v² — this limits the maximum rim speed of a flywheel (cast iron ≈ 30 m/s).

Example: A flywheel of mass 500 kg, radius of gyration 0.6 m, mean speed 300 rpm, maximum fluctuation of energy 5 kJ. Find C_s and max and min speeds.

Solution:

1. I = 500 × 0.36 = 180 kg·m² ; ω = 2π × 300/60 = 31.42 rad/s ; ω² = 987
2. $C_s = \Delta E/(I\omega^2) = 5000/(180 \times 987) =$ **0.0281 (2.81%)**
3. $N_{max} - N_{min} = 0.0281 \times 300 = 8.4$ rpm → $N_{max} =$ **304.2**, $N_{min} =$ **295.8 rpm**

1. Key Points

- **Turning moment diagram:** a graph of crank torque against crank angle. The area under the curve gives work done per cycle; the mean torque line divides it into areas above (+) and below (−).
- **Fluctuation of energy:** the difference between maximum and minimum energies, $\Delta E = E_{max} - E_{min}$, found from the areas of the turning moment diagram.
- **Coefficient of fluctuation of speed:** $C_s = (N_1 - N_2)/N$, where N is the mean speed. Its reciprocal is the **coefficient of steadiness** $m = 1/C_s$.
- **Coefficient of fluctuation of energy:** C_E = ΔE ÷ work done per cycle.
- **Energy stored:** $E = \frac{1}{2}I\omega^2$; maximum fluctuation $\Delta E = I\omega^2 C_s = mk^2\omega^2 C_s = 2EC_s$.
- **Rim-type flywheel:** about 90% of the moment of inertia comes from the rim; the arms and hub contribute only about 10%.
- **Hoop stress in the rim:** $\sigma = \rho v^2$, independent of the cross-section; hence the rim speed is limited (about 25–30 m/s for cast iron).
- **Flywheel vs governor:** a flywheel controls speed fluctuation within one cycle; a governor controls the mean speed over many cycles when load changes.

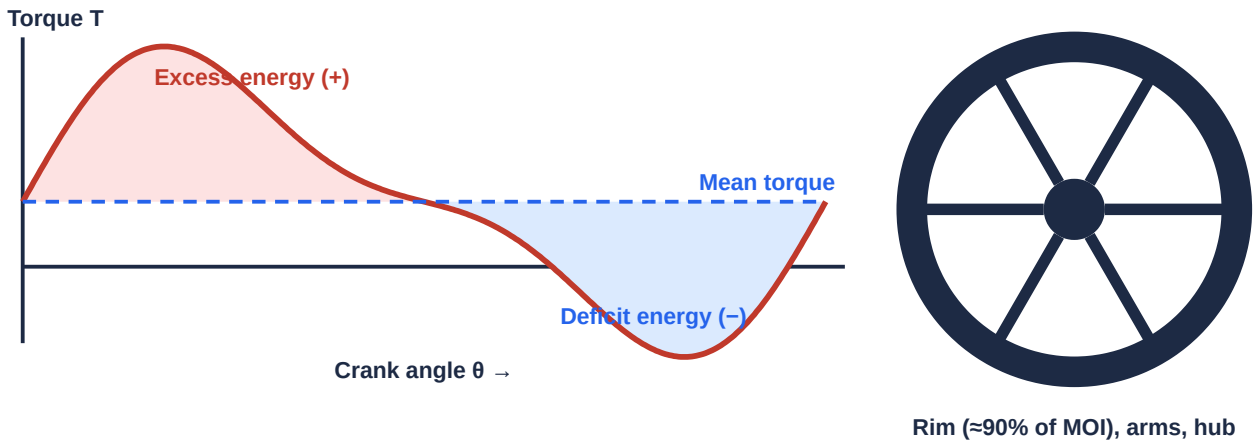


Fig 3.1: Turning moment diagram and rim-type flywheel

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Kinetic energy	$E = \frac{1}{2} I\omega^2$, $I = mk^2$
Solid disc	$I = \frac{1}{2} m r^2$
Rim (ring)	$I = m r^2$
C_s	$(N_1 - N_2)/N = (\omega_1 - \omega_2)/\omega$
ΔE	$I\omega^2 C_s = mk^2\omega^2 C_s = 2EC_s$
Hoop stress in rim	$\sigma = \rho v^2$

3. Detailed Explanation

- **Need for a flywheel:** in an I.C. engine, energy is produced only in the power stroke, while the load demands uniform energy. A four-stroke single-cylinder engine has one power stroke in 720° of crank rotation, so it needs the largest flywheel; multi-cylinder engines need smaller ones.
- **Punching press:** here the supply (motor) is uniform but the demand comes in short bursts. The flywheel stores energy during the idle period and releases it during punching, so a much smaller motor is sufficient.
- **Derivation of ΔE:** $\Delta E = \frac{1}{2} I(\omega_1^2 - \omega_2^2) = I \cdot \omega \cdot (\omega_1 - \omega_2) = I\omega^2 C_s$, taking $\omega = (\omega_1 + \omega_2)/2$.
- **Mass of flywheel:** for a given ΔE and C_s, $m = \Delta E/(k^2\omega^2 C_s)$. A higher speed or a larger radius of gyration allows a lighter flywheel.
- **Material and speed limit:** since $\sigma = \rho v^2$, the peripheral speed is limited by the material; cast iron is common for low speeds, steel for higher speeds.
- **Typical C_s values:** crushing machines about 0.2, punching presses 0.1, engines 0.02–0.03, electric generators 0.002–0.003.

4. Comparison Table — Flywheel vs Governor

Basis	Flywheel	Governor
Function	Reduces speed fluctuation within a cycle	Controls mean speed when load changes
Control of fuel supply	No	Yes, regulates fuel quantity
Time period	One cycle	Many cycles (long period)
Nature	Reservoir of energy	Controlling device

5. Solved Numerical Problems

Example 1. The maximum and minimum speeds of a flywheel are 210 rpm and 190 rpm. Find the coefficient of fluctuation of speed and the coefficient of steadiness.

Solution:

1. $N = (210 + 190)/2 = 200$ rpm
2. $C_s = (210 - 190)/200 =$ **0.1**
3. Coefficient of steadiness $m = 1/C_s =$ **10**

Example 2. A flywheel has mass 800 kg, radius of gyration 0.8 m, mean speed 250 rpm and $C_s = 0.02$. Find the maximum fluctuation of energy.

Solution:

1. $\omega = 2\pi \times 250/60 = 26.18$ rad/s, so $\omega^2 = 685.4$
2. $\Delta E = mk^2\omega^2C_s = 800 \times 0.64 \times 685.4 \times 0.02 =$ **7.02 kJ**

Example 3. $\Delta E = 20$ kJ, mean speed 300 rpm, $C_s = 0.02$ and radius of gyration 0.75 m. Find the mass of the flywheel required.

Solution:

1. $\omega = 31.42$ rad/s, so $\omega^2 = 986.96$
2. $m = \Delta E/(k^2\omega^2C_s) = 20000/(0.5625 \times 986.96 \times 0.02) = 20000/11.10 =$ **about 1801 kg**

Example 4. A cast iron rim ($\rho = 7200$ kg/m^3) has a peripheral velocity of 25 m/s. Find the hoop stress in the rim.

Solution:

1. $\sigma = \rho v^2 = 7200 \times 25^2 = 7200 \times 625 =$ **4.5 MPa**

6. Frequently Asked Facts

- $\Delta E = I\omega^2C_s = 2EC_s$; coefficient of steadiness $= 1/C_s$.
- Hoop stress in a flywheel rim $\sigma = \rho v^2$ – independent of the area of cross-section.
- A four-stroke single-cylinder engine needs the largest flywheel (one power stroke in 720°).

7. Note (Confusing Facts)

- A flywheel does NOT control the mean speed – that is the job of the governor.
- C_s uses the **mean** speed in the denominator, not the maximum speed.
- Kinetic energy varies as ω^2, so doubling the speed makes the energy four times.

Remember: One line: flywheel – fluctuation within a cycle; governor – mean speed over many cycles. Assertion–Reason: (A) A punching press with a flywheel needs a smaller motor. (R) The flywheel stores energy during the idle period and supplies it during punching. — Both true; R explains A.

30 Most Important One-Liner Facts — Flywheel & Fluctuation of Energy

1. Main function of a flywheel?

— To reduce speed fluctuation within a cycle.

2. A flywheel acts as?

— A reservoir of energy.

3. Turning moment diagram is a plot of?

— Crank torque vs crank angle.

4. Area under turning moment diagram gives?

— Work done per cycle.

5. Coefficient of fluctuation of speed?

— $C_s = (N_1 - N_2)/N$.

6. Coefficient of steadiness?

— $m = 1/C_s$.

7. Coefficient of fluctuation of energy?

— $\Delta E \div$ work done per cycle.

8. Kinetic energy of a flywheel?

— $E = \frac{1}{2}I\omega^2$.

9. Maximum fluctuation of energy?

— $\Delta E = I\omega^2 C_s$.

10. ΔE in terms of E?

— $\Delta E = 2EC_s$.

11. Moment of inertia of a rim?

— $I = mr^2$.

12. Moment of inertia of a solid disc?

— $I = \frac{1}{2}mr^2$.

13. Share of rim in total MOI of a rim-type flywheel?

— About 90%.

14. Hoop stress in a rotating rim?

— $\sigma = \rho v^2$.

15. Does hoop stress depend on cross-section?

— No.

16. Crank rotation in one cycle of a 4-stroke engine?

— 720°.

17. Engine needing the largest flywheel?

— Four-stroke single cylinder.

18. Effect of doubling speed on KE?

— Becomes four times.

19. Flywheel mass for given ΔE?

— $m = \Delta E/(k^2\omega^2 C_s)$.

20. Effect of higher speed on flywheel size?

— Smaller flywheel is needed.

21. Benefit of a flywheel in a punching press?

— A smaller motor can be used.

22. C_s in terms of angular speeds?

— $2(\omega_1 - \omega_2)/(\omega_1 + \omega_2)$.

23. Typical C_s for punching press?

— About 0.1.

24. Typical C_s for electric generators?

— About 0.002–0.003.

25. Common flywheel material for low speeds?

— Cast iron.

26. Does a flywheel control fuel supply?

— No.

27. Device that controls mean speed?

— Governor.

28. Flywheel works over what period?

— Within one cycle (continuously).

29. Governor works over what period?

— Many cycles (intermittently).

30. Which part stores most energy in a flywheel?

— The rim.

Practice MCQs — Flywheel & Fluctuation of Energy

Q1. The main function of a flywheel is:

(a) To control mean speed

(b) To reduce speed fluctuation within a cycle

(c) To control fuel

(d) To increase torque

✔ **Answer: (b) To reduce speed fluctuation within a cycle**

Explanation: A flywheel stores excess energy and releases it when needed, smoothing speed within a cycle.

Q2. If Cs = 0.05, the coefficient of steadiness is:

(a) 5

(b) 10

(c) 20

(d) 50

✔ **Answer: (c) 20**

Explanation: Coefficient of steadiness = $1/C_s = 1/0.05 = 20$.

Q3. The formula for ΔE is:

(a) $I\omega\ C_s$

(b) $I\omega^2\ C_s$

(c) $\frac{1}{2}I\omega\ C_s$

(d) $I\omega^2/C_s$

✔ **Answer: (b) $I\omega^2\ C_s$**

Explanation: $\Delta E = \frac{1}{2}I(\omega_1^2 - \omega_2^2) = I\omega^2 C_s$.

Q4. Hoop stress in a flywheel rim is:

(a) ρv

(b) ρv²

(c) ρ²v

(d) v²/ρ

✔ **Answer: (b) ρv²**

Explanation: Centrifugal hoop stress in a thin rim = ρv², independent of section.

Q5. In one cycle of a four-stroke engine, the crank rotates through:

(a) 180°

(b) 360°

(c) 540°

(d) 720°

✔ **Answer: (d) 720°**

Explanation: One cycle of a four-stroke engine takes two crank revolutions = 720°.

Q6. Which engine needs the largest flywheel?

(a) 4-stroke single cylinder
(b) 2-stroke single cylinder
(c) 4-stroke multi-cylinder
(d) 2-stroke multi-cylinder

✔ **Answer: (a) 4-stroke single cylinder**

Explanation: Only one power stroke in 720° gives the largest fluctuation, so the biggest flywheel.

Q7. In a rim-type flywheel, most of the moment of inertia comes from:

(a) Hub
(b) Arms
(c) Rim
(d) Shaft

✔ **Answer: (c) Rim**

Explanation: Mass is concentrated at the largest radius in the rim, giving about 90% of I.

Q8. If the speed is doubled, the KE of the flywheel becomes:

(a) Double
(b) Four times
(c) Half
(d) Unchanged

✔ **Answer: (b) Four times**

Explanation: $E = \frac{1}{2}I\omega^2$; doubling ω gives four times the energy.

Q9. Coefficient of fluctuation of energy is:

(a) ΔE ÷ work done per cycle
(b) E_{max}/E_{min}
(c) ΔE × work
(d) Work ÷ ΔE

✔ **Answer: (a) ΔE ÷ work done per cycle**

Explanation: C_E = maximum fluctuation of energy ÷ work done per cycle.

Q10. Using a flywheel in a punching press:

(a) Needs a bigger motor
(b) Allows a smaller motor
(c) Makes punching faster
(d) Has no benefit

✔ **Answer: (b) Allows a smaller motor**

Explanation: The flywheel supplies the peak energy for punching, so a smaller motor works.

SSC JE (Mechanical) Previous Year Questions — Flywheel & Fluctuation of Energy

Q1. A prime mover engine must have both flywheel and governor for:

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(a) Increasing speed
(b) Minimising speed fluctuations
(c) Saving fuel
(d) Increasing load

✔ **Answer: (b) Minimising speed fluctuations**

Explanation: The flywheel limits cyclic fluctuation and the governor maintains mean speed – both together minimise speed fluctuations.

Q2. Which is used for controlling speed fluctuations in an engine? SSC JE 2018 (22 January, Evening)

(a) Control valve
(b) Flywheel
(c) Governor
(d) Both flywheel and governor

✔ **Answer: (d) Both flywheel and governor**

Explanation: Within-cycle fluctuation is handled by the flywheel and load-change fluctuation by the governor.

Q3. Coefficient of fluctuation of speed in terms of angular speeds: SSC JE 2018 (22 January, Evening)

(a) $(\omega_1 - \omega_2)/(\omega_1 + \omega_2)$
(b) $2(\omega_1 - \omega_2)/(\omega_1 + \omega_2)$
(c) $(\omega_1 + \omega_2)/2(\omega_1 - \omega_2)$
(d) $(\omega_1 - \omega_2)/2\omega_1$

✔ **Answer: (b) $2(\omega_1 - \omega_2)/(\omega_1 + \omega_2)$**

Explanation: $C_s = (\omega_1 - \omega_2)/\omega$ with $\omega = (\omega_1 + \omega_2)/2$, giving $2(\omega_1 - \omega_2)/(\omega_1 + \omega_2)$.

Q4. A flywheel ($I = 2500$ kg·m²) starts from rest with $\alpha = 0.6$ rad/s². KE after 10 s (kN·m): SSC JE 2018 (23 January, Evening)

(a) 45
(b) 450
(c) 4.5
(d) 4500

✔ **Answer: (a) 45**

Explanation: $\omega = \alpha t = 6$ rad/s; KE = ½ × 2500 × 36 = 45000 J = 45 kN·m.

Q5. Correct statement about operation of flywheel and governor: SSC JE 2023 (11 October, Shift-2)

(a) Both are intermittent
(b) Governor continuous, flywheel intermittent
(c) Both are continuous
(d) Flywheel continuous, governor more or less intermittent

✔ **Answer: (d) Flywheel continuous, governor more or less intermittent**

Explanation: The flywheel acts continuously in every cycle; the governor acts only when the load changes.

UPSSSC ALERT

Chapter 4: Power Transmission by Belts (Flat & V-belt)

Power Transmission by Belts (Flat & V-belt) – Basic Concept

Belts, ropes and chains transmit power between shafts that are some distance apart. A belt drive works by **friction** between the belt and the pulley. Flat belts are used for moderate power over larger centre distances, while V-belts give a higher grip and are used for shorter centre distances and higher power.

Understand in Detail

4.1 Belt drive — when and why

When two shafts are fairly far apart (like the motor and the stones of a flour mill, or line-shafting in old factories), gears are costly and difficult. Then **pulleys** are fitted on both shafts and a **belt** is put over them — the **friction** between belt and pulley transmits the power. Advantages: cheap, quiet, absorbs shocks, slips under overload and protects the machine. Disadvantage: due to **slip** the velocity ratio is not exact.

Types: flat belt — large distance (up to ≈ 10 m), moderate power; **V-belt** — trapezoidal section, wedges into the pulley groove, short distance, more power, less slip; **circular (rope)** — very large distance; **timing (toothed) belt** — no slip, exact ratio (car engine camshaft). **Open belt** — both pulleys rotate in the same direction; **crossed belt** — in opposite directions.

Derivation: Velocity ratio and slip

1. Belt speed = peripheral speed of pulley ; with no slip $\pi\, d_1\, N_1 = \pi\, d_2\, N_2 \rightarrow \mathbf{N_2/N_1 = d_1/d_2}$
2. With belt thickness t: $N_2/N_1 = (d_1 + t)/(d_2 + t)$
3. With total slip s%: $N_2/N_1 = (d_1/d_2)(1 - s/100)$
4. **Creep** — the small relative motion caused by the change in elastic stretch of the belt as it passes from tight to slack side ; it is different from slip

4.2 Belt length, tension ratio and maximum power

Of a belt wrapped on pulleys one side is tighter (**tight side, T_1** — the driver pulls it) and the other slack (**slack side, T_2**). Their difference turns the driven pulley.

Derivation: Main formulae

1. **Open belt length:** $L = \pi(r_1 + r_2) + 2x + (r_1 - r_2)^2/x$; **crossed:** $L = \pi(r_1 + r_2) + 2x + (r_1 + r_2)^2/x$ (x = centre distance) — crossed-belt length depends only on $(r_1 + r_2)$
2. **Tension ratio (flat):** taking equilibrium of a small belt element and integrating → $\mathbf{T_1/T_2 = e^{\mu\theta}}$ (θ in radians, angle of wrap on the smaller pulley)
3. **V-belt:** with groove half-angle β the wedge action increases the effective μ → $\mathbf{T_1/T_2 = e^{\mu\theta/\sin\beta}}$ — that is why a V-belt transmits more power
4. **Power $\mathbf{P = (T_1 - T_2)\,v}$** ; centrifugal tension $\mathbf{T_c = m\,v^2}$ (m = mass per metre) ; maximum tension $T = T_1 + T_c = \sigma \times b \times t$
5. **Condition for maximum power:** $dP/dv = 0$ → $\mathbf{T_c = T/3}$; belt velocity $\mathbf{v = \sqrt{(T/3m)}}$
6. **Initial tension:** $T_0 = (T_1 + T_2)/2$ (centrifugal neglected) — the tension given while mounting the belt

Example: A flat belt with 160° wrap, $\mu = 0.3$, maximum allowable tension 2000 N, speed 15 m/s. Power? (T_c negligible)

Solution:

1. $\theta = 160 \times \pi/180 = 2.793$ rad ; $e^{0.3 \times 2.793} = e^{0.838} = 2.311$
2. $T_1 = 2000$ → $T_2 = 2000/2.311 = 865.4$ N
3. $P = (2000 - 865.4) \times 15 = 17019$ W ≈ **17 kW**

1. Key Points

- **Open belt drive:** both shafts rotate in the **same direction**. **Crossed belt drive:** the shafts rotate in **opposite directions**; it has a larger angle of contact but more wear.
- **Velocity ratio:** $N_2/N_1 = d_1/d_2$. With belt thickness t: $N_2/N_1 = (d_1 + t)/(d_2 + t)$. With total slip s%: multiply by $(1 - s/100)$.
- **Slip:** relative motion between belt and pulley due to insufficient friction. **Creep:** the slow relative movement caused by unequal stretching of the belt on the tight and slack sides (due to belt elasticity).
- **Tension ratio:** flat belt $T_1/T_2 = e^{\mu\theta}$; V-belt or rope $T_1/T_2 = e^{\mu\theta\ \mathrm{cosec}\ \beta}$, where β is the half groove angle.
- **Power transmitted:** $P = (T_1 - T_2)v$, where T_1 is the tight-side and T_2 the slack-side tension.
- **Centrifugal tension:** $T_c = mv^2$ (m = mass per metre). It acts equally on both sides and does **not** affect the power transmitted, but it reduces the useful tension.
- **Condition for maximum power:** $T_{max} = 3T_c$, i.e. $v = \sqrt{(T_{max}/3m)}$.
- **Initial tension:** $T_0 = (T_1 + T_2)/2$ (neglecting centrifugal tension).

- **Design pulley:** the belt slips first on the **smaller** pulley (smaller angle of contact), so the design is based on the smaller pulley.
- **Idler (jockey) pulley:** used to increase the angle of contact and hence the power transmitted; stepped cone pulleys use an open or crossed belt to give different speeds.

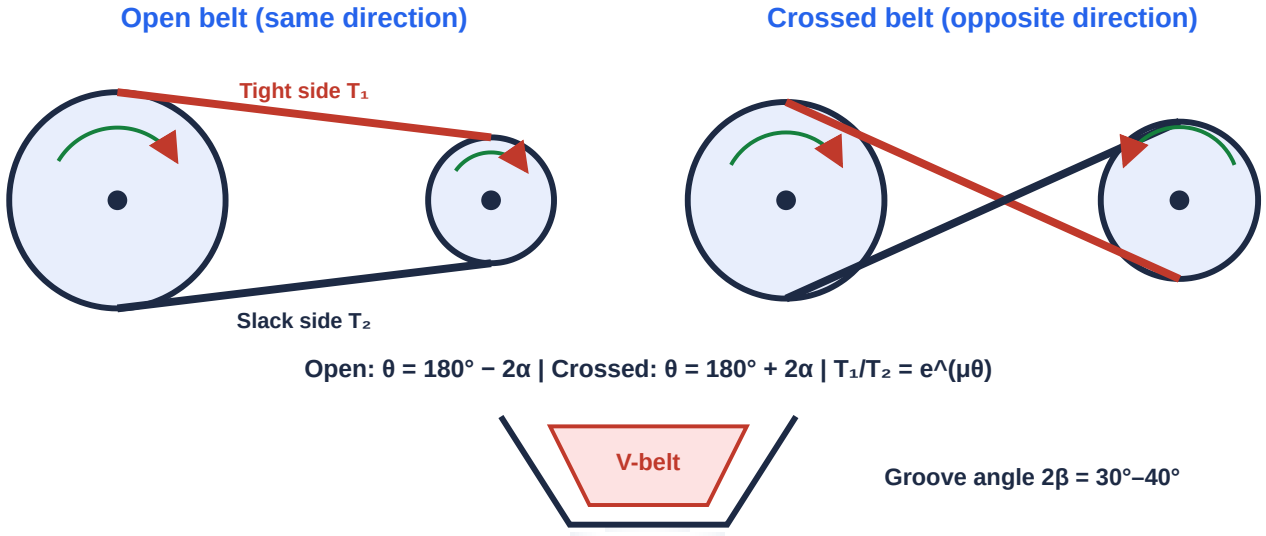


Fig 4.1: Open and crossed belt drives and V-belt groove

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Length of open belt	$L = \pi(r_1+r_2) + 2x + (r_1-r_2)^2/x$
Length of crossed belt	$L = \pi(r_1+r_2) + 2x + (r_1+r_2)^2/x$
Flat belt	$T_1/T_2 = e^{\mu\theta}$
V-belt / rope	$T_1/T_2 = e^{\mu\theta \operatorname{cosec} \beta}$
Centrifugal tension	$T_c = mv^2$
Maximum power	$T_{max} = 3T_c$, $v = \sqrt{(T/3m)}$
Initial tension	$T_0 = (T_1 + T_2)/2$

3. Detailed Explanation

- **Angle of contact:** for an open belt $\theta = 180° - 2\alpha$ on the smaller pulley, where $\sin\alpha = (r_1 - r_2)/x$. For a crossed belt $\theta = 180° + 2\alpha$ with $\sin\alpha = (r_1 + r_2)/x$, and it is the same on both pulleys.
- **Length of crossed belt** depends on the sum $(r_1 + r_2)$; hence for stepped cone pulleys with a crossed belt the same belt fits all steps if the sum of radii is constant.
- **Why V-belts grip more:** the wedge action in the groove increases the normal reaction, so the effective coefficient of friction becomes $\mu \operatorname{cosec} \beta$. With a groove angle of 30°–40°, the tension ratio is much higher than for a flat belt.

- **Effect of centrifugal tension:** total tight-side tension $T_{t1} = T_1 + T_c$. Since T_c is added to both sides, $(T_1 - T_2)$ is unchanged and so is the power, but the maximum safe tension available for power is reduced.
- **Belt materials:** leather, cotton/fabric, rubber, balata; V-belts are made of rubber with fabric and cord reinforcement and are endless.
- **Chain drives** give a positive (no slip) drive, while belts may slip; ropes are used for very long distances and heavy power (e.g. cranes, mines).

4. Comparison Table — Flat Belt vs V-belt

Basis	Flat Belt	V-belt
Tension ratio	$e^{\mu\theta}$	$e^{\mu\theta \operatorname{cosec} \beta}$ (higher)
Slip	More	Very little
Centre distance	Large (shafts far apart)	Small (shafts close)
Crossed drive	Possible	Generally not
Power transmitted	Less	More

5. Solved Numerical Problems

Example 1. A driving pulley of 600 mm diameter runs at 300 rpm and drives a pulley of 300 mm diameter. Find the speed of the driven pulley: (i) normally, (ii) taking belt thickness 5 mm, (iii) with a total slip of 2%.

Solution:

1. (i) $N_2 = 300 \times 600/300 =$ **600 rpm**
2. (ii) $N_2 = 300 \times (600 + 5)/(300 + 5) =$ **595.1 rpm**
3. (iii) $N_2 = 600 \times (1 - 0.02) =$ **588 rpm**

Example 2. Two pulleys have radii 300 mm and 150 mm and a centre distance of 2 m. Find the lengths of open and crossed belts.

Solution:

1. Open belt: $L = \pi(0.45) + 2 \times 2 + (0.15)^2/2 = 1.414 + 4 + 0.011 =$ **5.425 m**
2. Crossed belt: $L = \pi(0.45) + 4 + (0.45)^2/2 = 1.414 + 4 + 0.101 =$ **5.515 m**

Example 3. In a flat belt $T_1 = 2000$ N, $\mu = 0.3$, angle of contact 165° and belt speed 15 m/s. Find the power transmitted.

Solution:

1. $\theta = 165 \times \pi/180 = 2.88$ rad, so $e^{\mu\theta} = e^{0.864} = 2.373$
2. $T_2 = 2000/2.373 = 843$ N
3. $P = (2000 - 843) \times 15 =$ **17.36 kW**

Example 4. A belt has a mass of 0.6 kg/m and a maximum permissible tension of 1200 N. Find the belt speed for maximum power and the centrifugal tension.

Solution:

1. $v = \sqrt{(T/3m)} = \sqrt{(1200/1.8)} =$ **25.8 m/s**
2. $T_c = T_{max}/3 =$ **400 N**

6. Frequently Asked Facts

- Maximum power: $T_{max} = 3T_c$, $v = \sqrt{(T/3m)}$.
- Centrifugal tension does not affect the power transmitted.
- Creep is due to the elasticity of the belt; slip is due to insufficient friction.

7. Note (Confusing Facts)

- Crossed belt – opposite direction; open belt – same direction.
- The design is based on the **smaller** pulley, because the belt slips there first.
- The angle in $e^{\mu\theta}$ must be in **radians**, not degrees.

Remember: One line: a V-belt grips better because its effective friction is μ cosec β. Assertion–Reason: (A) Centrifugal tension does not change the power transmitted. (R) It acts equally on the tight and slack sides. — Both true; R explains A.

30 Most Important One-Liner Facts — Power Transmission by Belts (Flat & V-belt)

1. Direction of rotation in an open belt drive?

— Same direction.

2. Direction of rotation in a crossed belt drive?

— Opposite directions.

3. Velocity ratio of a belt drive?

— $N_2/N_1 = d_1/d_2$.

4. VR with belt thickness t?

— $(d_1 + t)/(d_2 + t)$.

5. Effect of slip on driven speed?

— Reduces it by factor $(1 - s/100)$.

6. Cause of creep?

— Elasticity (unequal stretching) of the belt.

7. Cause of slip?

— Insufficient friction between belt and pulley.

8. Tension ratio of a flat belt?

— $e^{\mu\theta}$.

9. Tension ratio of a V-belt?

— $e^{\mu\theta \operatorname{cosec} \beta}$.

10. Usual groove angle of a V-belt?

— 30°–40°.

11. Power transmitted by a belt?

— $P = (T_1 - T_2)v$.

12. Centrifugal tension?

— $T_c = mv^2$.

13. Effect of T_c on power?

— No effect.

14. Condition for maximum power?

— $T_{max} = 3T_c$.

15. Belt speed for maximum power?

— $v = \sqrt{(T_{max}/3m)}$.

16. Initial tension?

— $T_0 = (T_1 + T_2)/2$.

17. Design of a belt drive is based on?

— The smaller pulley.

18. Purpose of an idler pulley?

— To increase the angle of contact.

19. Length of an open belt depends on?

— $(r_1 - r_2)$.

20. Length of a crossed belt depends on?

— $(r_1 + r_2)$.

21. Belt suitable for stepped cone pulleys?

— Crossed belt.

22. Angle of contact larger in which drive?

— Crossed belt.

23. Unit of θ in tension ratio?

— Radian.

24. Drive with no slip?

— Chain or gear drive (positive drive).

25. Drive for very long distances and heavy loads?

— Rope drive.

26. Is a V-belt suitable for crossed drive?

— Generally no.

27. Material of V-belts?

— Rubber with fabric and cord reinforcement.

28. Flat belt is suitable for?

— Large centre distances.

29. V-belt is suitable for?

— Short centre distances, higher power.

30. Tight side of a belt carries?

— Higher tension T_1.

Practice MCQs — Power Transmission by Belts (Flat & V-belt)

Q1. In a crossed belt drive, the pulleys rotate:

(a) In the same direction

(b) In opposite directions

(c) They stop

(d) No fixed rule

✔ **Answer: (b) In opposite directions**

Explanation: Crossing the belt reverses the direction of the driven pulley.

Q2. Tension ratio of a flat belt is:

(a) $\mu\theta$

(b) $e^{\mu\theta}$

(c) $e^{\theta/\mu}$

(d) μ/θ

✔ **Answer: (b) $e^{\mu\theta}$**

Explanation: For a flat belt, $T_1/T_2 = e^{\mu\theta}$ with θ in radians.

Q3. At maximum power, Tc equals:

(a) $T_{max}/2$

(b) $T_{max}/3$

(c) T_{max}

(d) $2T_{max}/3$

✔ **Answer: (b) $T_{max}/3$**

Explanation: Maximum power occurs when $T_c = T_{max}/3$.

Q4. Centrifugal tension affects the transmitted power by:

(a) Increasing it
(b) Decreasing it
(c) Not affecting it
(d) Doubling it

✔ **Answer: (c) Not affecting it**

Explanation: T_c adds equally to both sides, so $(T_1 - T_2)$ and the power are unchanged.

Q5. Groove angle of a V-belt is generally:

(a) 10°–20°
(b) 30°–40°
(c) 50°–60°
(d) 90°

✔ **Answer: (b) 30°–40°**

Explanation: Standard V-belt groove angle is 30°–40°.

Q6. Creep in a belt is caused by:

(a) Low friction
(b) Elasticity of belt
(c) Pulley crowning
(d) Belt thickness

✔ **Answer: (b) Elasticity of belt**

Explanation: Creep is due to the belt's elasticity – unequal stretching on the two sides.

Q7. Which belt is suitable for stepped cone pulleys?

(a) Open
(b) Crossed
(c) V-belt
(d) Rope

✔ **Answer: (b) Crossed**

Explanation: For a crossed belt, length depends on $(r_1 + r_2)$, which stays constant for stepped cones.

Q8. An idler pulley is used to:

(a) Increase speed
(b) Increase angle of contact
(c) Shorten the belt
(d) Increase slip

✔ **Answer: (b) Increase angle of contact**

Explanation: An idler pulley increases the angle of contact on the smaller pulley.

Q9. Velocity ratio considering belt thickness:

(a) d_1/d_2
(b) $(d_1 + t)/(d_2 + t)$
(c) $(d_1 - t)/(d_2 - t)$
(d) d_2/d_1

✔ **Answer: (b) $(d_1 + t)/(d_2 + t)$**

Explanation: Taking the belt thickness, the effective diameters become (d + t).

Q10. A belt drive is designed on the basis of which pulley?

(a) Larger
(b) Smaller
(c) Both
(d) Idler

✔ **Answer: (b) Smaller**

Explanation: The smaller pulley has the smaller angle of contact, so slip begins there – design is based on it.

SSC JE (Mechanical) Previous Year Questions — Power Transmission by Belts (Flat & V-belt)

Q1. Velocity ratio of a belt drive with slip s% and thickness t:

SSC JE 2018 (22 January, Evening)

(a) $N_2/N_1 = (d_1 + t)/(d_2 + t) \times (1 - s/100)$

(b) $N_2/N_1 = (d_1 - t)/(d_2 - t) \times (1 - s/100)$

(c) $N_2/N_1 = (d_1 + t)/(d_2 + t) \times (1 + s/100)$

(d) $N_2/N_1 = d_2/d_1 \times (1 - s/100)$

✔ **Answer: (a) $N_2/N_1 = (d_1 + t)/(d_2 + t) \times (1 - s/100)$**

Explanation: Thickness gives effective diameters (d + t); slip reduces speed by the factor (1 − s/100).

Q2. Creep in a belt occurs due to:

SSC JE (Previous Year)

(a) Uneven contraction and extension of belt

(b) Weak pulley material

(c) Weak belt material

(d) Improper crowning

✔ **Answer: (a) Uneven contraction and extension of belt**

Explanation: Creep results from uneven extension and contraction of the belt due to varying tension.

UPSSSC ALERT

Chapter 5: Clutches (Plate & Conical)

Clutches (Plate & Conical) – Basic Concept

A **clutch** is a device used to connect or disconnect the driving shaft (engine) and the driven shaft (gear box) at the will of the operator, while the engine is running. Friction clutches transmit torque through the friction between contact surfaces pressed together by springs.

Understand in Detail

5.1 Clutch — the 'switch' between engine and wheels

Why do we press the clutch while changing gears on a motorcycle? The engine keeps rotating; to change gear or stop the vehicle the engine must be **temporarily disconnected** from the wheels and then **reconnected** gradually, without a jerk. That is the job of a **clutch** — a device to connect and disconnect two coaxial shafts at will.

Types: positive clutch (jaw/dog) — teeth interlock, no slip, but cannot be engaged while running; **friction clutch** — by friction between two surfaces, engages gradually: **single-plate** (cars), **multi-plate** (motorcycles — more torque in less space), **cone**, **centrifugal** (mopeds — engages automatically as speed rises). In a plate clutch **springs** apply the pressure and pressing the pedal separates the plates.

5.2 Friction torque — uniform pressure and uniform wear

A clutch plate is an annular ring — outer radius R_1, inner R_2. The question is how the axial force W is distributed over the plate. There are two theories:

Uniform pressure: in a new, perfectly flat clutch the pressure p is the same everywhere. **Uniform wear:** in use the outer portion (higher rubbing speed) wears more, until **p × r = constant** — an old clutch reaches this state. This theory gives a somewhat **smaller** torque, so it is used in design (**the safer side**).

Derivation: Friction torque (one friction surface)

1. Ring of width dr at radius r: force $dW = p\,2\pi r\,dr$; friction torque $dT = \mu\,dW\,r$
2. **Uniform pressure:** $p = W/[\pi(R_1^2 - R_2^2)] \rightarrow$ **$T = (2/3)\,\mu\,W\,(R_1^3 - R_2^3)/(R_1^2 - R_2^2)$**
3. **Uniform wear:** $p\,r = C \rightarrow W = 2\pi C(R_1 - R_2) \rightarrow$ **$T = \mu\,W\,(R_1 + R_2)/2$** — i.e. at the mean radius
4. **n friction surfaces:** total T = n × (T of one surface) ; a single-plate clutch has friction on both sides → n = 2 ; multi-plate $n = n_1 + n_2 - 1$
5. **Cone clutch** (semi-cone angle α): $T = \mu\,W\,R_m/\sin\alpha$ — smaller α gives larger torque, but a very small α may make the clutch stick ($\alpha \approx 12.5°$)
6. **Power $P = 2\pi NT/60$**

Example: Single-plate clutch (friction on both sides), outer dia 300 mm, inner 200 mm, μ = 0.3, axial force 4 kN. Power at 1500 rpm on uniform wear?

Solution:

1. $R_m = (150 + 100)/2 = 125$ mm = 0.125 m
2. $T = n \mu W R_m = 2 \times 0.3 \times 4000 \times 0.125 =$ **300 N·m**
3. $P = 2\pi \times 1500 \times 300/60 = 47124$ W ≈ **47.1 kW**

1. Key Points

- **Types:** positive clutches (jaw/claw clutch – no slip) and friction clutches (single plate, multi-plate, cone, centrifugal).
- **Single plate clutch:** used in cars; the friction plate has lining on both sides, so n = 2 effective surfaces.
- **Multi-plate clutch:** used in scooters and motorcycles where space is limited; number of contact pairs $n = n_1 + n_2 - 1$.
- **Uniform pressure theory:** applicable to a new clutch; p = constant over the whole surface.
- **Uniform wear theory:** applicable to a used (worn) clutch; p × r = constant. Clutches are **designed on uniform wear** because it gives a lower (safer) torque.
- **Cone clutch:** T = μWR cosec α, where α is the semi-cone angle (usually 12°–15°). A smaller cone angle gives more torque for the same axial force.
- **Centrifugal clutch:** engages automatically when speed increases; used in mopeds.

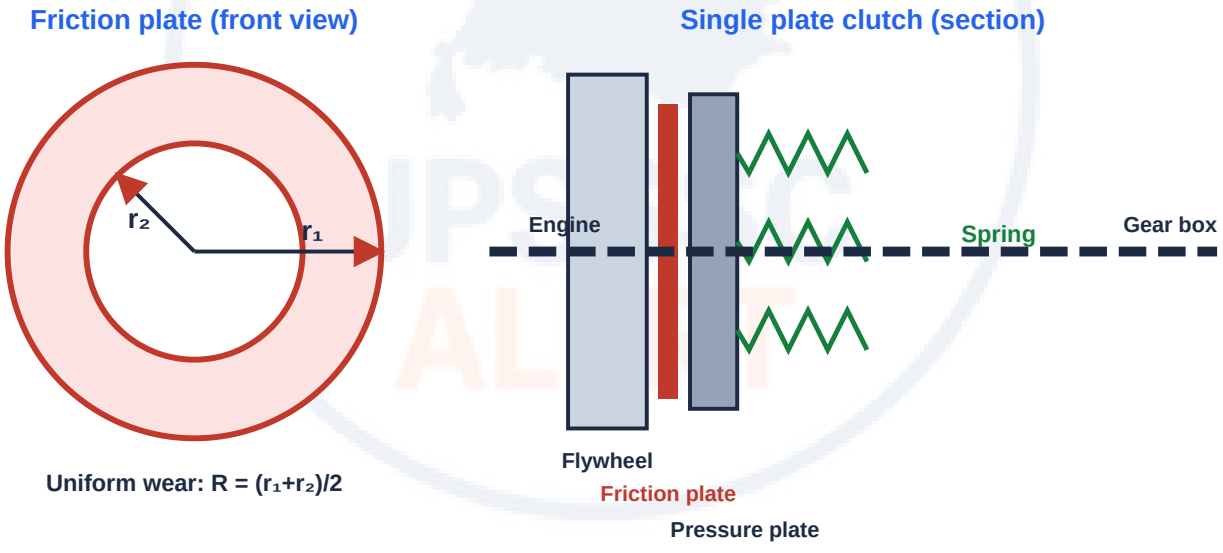


Fig 5.1: Clutch plate radii and simple section of a single plate clutch

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Torque (general)	$T = n \mu W R$
Uniform pressure – mean radius	$R = \frac{2}{3} (r_1^3 - r_2^3)/(r_1^2 - r_2^2)$
Uniform wear – mean radius	$R = (r_1 + r_2)/2$
Multi-plate clutch	$n = n_1 + n_2 - 1$
Cone clutch	$T = \mu W R \operatorname{cosec} \alpha$
Condition of uniform wear	$p \times r = \text{constant}$

3. Detailed Explanation

- **Uniform wear reasoning:** wear is proportional to the work of friction, i.e. to p × v. Since v ∝ r, uniform wear requires p × r = constant; therefore the pressure is **maximum at the inner radius**.
- **Axial load under uniform wear:** $W = 2\pi C(r_1 - r_2)$, where $C = p_{max} \times r_2$.
- **Uniform pressure gives a slightly larger torque** because its mean radius is slightly larger; designing on uniform wear is therefore on the safe side.
- **Plate clutch vs flat collar bearing:** the friction torque formulas are identical – a disc clutch is analysed exactly like a flat collar bearing.
- **Dry and wet clutches:** wet clutches run in oil, so their μ is lower and torque capacity is lower for the same size, but they have better cooling and longer life.
- **Friction lining materials:** asbestos-based (earlier), now non-asbestos organic, sintered metal and ceramic linings.

4. Comparison Table — Uniform Pressure vs Uniform Wear

Basis	Uniform Pressure	Uniform Wear
Applies to	New clutch	Old (worn) clutch
Condition	p = constant	p × r = constant
Mean radius	$\frac{2}{3}(r_1^3 - r_2^3)/(r_1^2 - r_2^2)$	$(r_1 + r_2)/2$
Torque	Slightly more	Slightly less (design basis)
Maximum pressure	Same everywhere	At inner radius

5. Solved Numerical Problems

Example 1. A single plate clutch (both sides effective) has $r_1 = 125$ mm, $r_2 = 75$ mm, $\mu = 0.25$ and axial load $W = 3$ kN. Find the torque by both theories.

Solution:

1. Uniform wear: $R = (0.125 + 0.075)/2 = 0.1$ m, so $T = 2 \times 0.25 \times 3000 \times 0.1 =$ **150 N·m**
2. Uniform pressure: $R = \frac{2}{3} \times (0.001953 - 0.000422)/(0.015625 - 0.005625) = 0.1021$ m, so $T =$ **153.1 N·m**

Example 2. A multi-plate clutch has 4 driving and 3 driven plates. $\mu = 0.3$, $W = 2$ kN and mean radius 60 mm. Find the torque and the power transmitted at 2000 rpm.

Solution:

1. $n = 4 + 3 - 1 = 6$
2. $T = 6 \times 0.3 \times 2000 \times 0.06 =$ **216 N·m**
3. $P = T\omega = 216 \times (2\pi \times 2000/60) = 216 \times 209.4 =$ **45.2 kW**

Example 3. A cone clutch has semi-cone angle 12.5°, $\mu = 0.25$, axial load 1 kN and mean radius 100 mm. Find the torque.

Solution:

1. $\operatorname{cosec} 12.5° = 1/0.2164 = 4.62$
2. $T = \mu W R \operatorname{cosec} \alpha = 0.25 \times 1000 \times 0.1 \times 4.62 =$ **115.5 N·m**

Example 4. For the clutch of Example 1, find the maximum pressure under uniform wear.

Solution:

1. $W = 2\pi p_{max} r_2 (r_1 - r_2)$, so $p_{max} = 3000/(2\pi \times 0.075 \times 0.05)$
2. $p_{max} = 3000/0.02356 =$ **127.3 kPa** (at the inner radius)

6. Frequently Asked Facts

- Clutches are designed on uniform wear: $R = (r_1 + r_2)/2$.
- Multi-plate: $n = n_1 + n_2 - 1$; single plate with two linings: $n = 2$.
- Cone clutch torque contains $\operatorname{cosec} \alpha$.

7. Note (Confusing Facts)

- Under uniform wear the maximum pressure is at the **inner** radius, not the outer.
- Uniform pressure gives more torque, but design uses uniform wear (lower, safer torque).
- A jaw clutch is a positive clutch – it cannot be engaged at high relative speed.

Remember: One line: new clutch – uniform pressure; old clutch – uniform wear. Assertion–Reason: (A) A cone clutch transmits more torque than a flat plate of the same size and load. (R) The wedge action multiplies the normal force by $\operatorname{cosec} \alpha$. — Both true; R explains A.

30 Most Important One-Liner Facts — Clutches (Plate & Conical)

1. Purpose of a clutch?

— To connect or disconnect engine and gear box.

2. Clutch used in cars?

— Single plate clutch.

3. Clutch used in motorcycles/scooters?

— Multi-plate clutch.

4. Clutch used in mopeds?

— Centrifugal clutch.

5. Example of a positive clutch?

— Jaw (claw) clutch.

6. General torque formula of a plate clutch?

— $T = n\mu WR$.

7. Effective surfaces in a single plate clutch with two linings?

— $n = 2$.

8. Contact pairs in a multi-plate clutch?

— $n = n_1 + n_2 - 1$.

9. Uniform pressure theory applies to?

— New clutch.

10. Uniform wear theory applies to?

— Old (worn) clutch.

11. Condition of uniform wear?

— $p \times r$ = constant.

12. Mean radius under uniform wear?

— $(r_1 + r_2)/2$.

13. Mean radius under uniform pressure?

— $\frac{2}{3}(r_1^3 - r_2^3)/(r_1^2 - r_2^2)$.

14. Theory used for design of clutches?

— Uniform wear.

15. Which theory gives higher torque?

— Uniform pressure.

16. Location of maximum pressure under uniform wear?

— Inner radius.

17. Torque of a cone clutch?

— $T = \mu WR \operatorname{cosec} \alpha$.

18. Usual semi-cone angle of a cone clutch?

— 12°–15°.

19. Effect of reducing cone angle?

— Torque increases.

20. Axial load under uniform wear?

— $W = 2\pi p_{max} r_2(r_1 - r_2)$.

21. Plate clutch torque is similar to?

— Flat collar bearing.

22. Why is wet clutch torque lower?

— Oil reduces μ.

23. Advantage of wet clutch?

— Better cooling and longer life.

24. Power transmitted by a clutch?

— $P = T\omega$.

25. Wear is proportional to?

— $p \times v$ ($p \times r$).

26. Force that engages a plate clutch?

— Spring force.

27. Clutch that engages automatically with speed?

— Centrifugal clutch.

28. Clutch with no slip?

— Positive (jaw) clutch.

29. Drawback of a jaw clutch?

— Cannot engage at high relative speed.

30. Modern friction lining materials?

— Non-asbestos organic, sintered metal, ceramic.

Practice MCQs — Clutches (Plate & Conical)

Q1. Which clutch is generally used in a car?

(a) Cone
(b) Single plate
(c) Centrifugal
(d) Jaw

✔ **Answer: (b) Single plate**

Explanation: Cars use a single plate dry clutch with lining on both sides.

Q2. Mean radius under uniform wear is:

(a) $(r_1 + r_2)/2$
(b) $(r_1 - r_2)/2$
(c) $\sqrt{(r_1 r_2)}$
(d) $\frac{2}{3}\, r_1$

✔ **Answer: (a) $(r_1 + r_2)/2$**

Explanation: Under uniform wear the mean radius is the arithmetic mean $(r_1 + r_2)/2$.

Q3. Condition of uniform wear is:

(a) p = constant
(b) p × r = constant
(c) p/r = constant
(d) p × r^2 = constant

✔ **Answer: (b) p × r = constant**

Explanation: Wear ∝ p × v and v ∝ r, so p × r = constant.

Q4. With 5 driving and 4 driven plates, the number of contact pairs is:

(a) 9
(b) 8
(c) 10
(d) 20

✔ **Answer: (b) 8**

Explanation: $n = n_1 + n_2 - 1 = 5 + 4 - 1 = 8$.

Q5. Torque of a cone clutch contains the factor:

(a) sin α
(b) cos α
(c) cosec α
(d) tan α

✔ **Answer: (c) cosec α**

Explanation: The wedge action multiplies the normal force by 1/sin α, giving cosec α.

Q6. Clutches are designed on which theory?

(a) Uniform pressure
(b) Uniform wear
(c) Average of both
(d) None

✔ **Answer: (b) Uniform wear**

Explanation: Uniform wear gives a slightly lower torque, so design on it is safe.

Q7. Under uniform wear, maximum pressure occurs at:

(a) Outer radius
(b) Inner radius
(c) Mean radius
(d) Uniform everywhere

✔ **Answer: (b) Inner radius**

Explanation: Since p × r = constant, p is largest where r is smallest – the inner radius.

Q8. The clutch used in mopeds is:

(a) Centrifugal
(b) Cone
(c) Jaw
(d) Single plate

✔ **Answer: (a) Centrifugal**

Explanation: Mopeds use a centrifugal clutch that engages automatically with speed.

Q9. An example of a positive clutch is:

(a) Jaw clutch
(b) Cone clutch
(c) Plate clutch
(d) Centrifugal clutch

✔ **Answer: (a) Jaw clutch**

Explanation: A jaw clutch locks positively by interlocking jaws – no slip.

Q10. In a single plate clutch with lining on both sides, n is:

(a) 1
(b) 2
(c) 3
(d) 4

✔ **Answer: (b) 2**

Explanation: Both faces of the plate are lined, so there are two friction surfaces.

SSC JE (Mechanical) Previous Year Questions — Clutches (Plate & Conical)

Q1. Which statement about dry and wet clutches is INCORRECT?

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(a) Heat dissipation is more difficult in dry clutch
(b) Wear is less in wet clutch
(c) Torque capacity of dry clutch is less than wet clutch
(d) Engagement is smoother in wet clutch

✔ **Answer: (c) Torque capacity of dry clutch is less than wet clutch**

Explanation: Wet clutches have lower μ, so the dry clutch has the higher torque capacity – the given statement is incorrect.

Q2. Torque capacity of a cone clutch by uniform wear theory:

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(a) $M_t = (\mu P/4 \sin \alpha)(D + d)$
(b) $M_t = (\mu P/4 \sin \alpha)(D - d)$
(c) $M_t = (\mu P/\sin \alpha)(D - d)$
(d) $M_t = (2\mu P/\sin \alpha)(D - d)$

✔ **Answer: (a) $M_t = (\mu P/4 \sin \alpha)(D + d)$**

Explanation: With uniform wear $R = (D + d)/4$ and $T = \mu PR/\sin \alpha = (\mu P/4 \sin \alpha)(D + d)$.

Q3. Frictional torque of a disc/plate clutch is equivalent to that of a: SSC JE 2018 (22 January, Evening)

(a) Flat pivot bearing
(b) Flat collar bearing
(c) Conical pivot bearing
(d) Trapezoidal pivot bearing

✔ **Answer: (b) Flat collar bearing**

Explanation: The friction torque of a disc clutch is calculated exactly like a flat collar bearing.

UPSSSC ALERT

Chapter 6: Gears, Types, Profiles & Gear Ratio

Gears, Types, Profiles & Gear Ratio – Basic Concept

Gears are toothed wheels that transmit motion and power between shafts by the successive engagement of teeth. Unlike belts they give a **positive drive** with an exact velocity ratio. They are used where the centre distance is small and a constant velocity ratio is required.

Understand in Detail

6.1 Gears — exact power transmission without slip

A belt slips, so where an **exact velocity ratio** is required (clocks, car gear-box, machine tools) **gears** are used — toothed wheels whose teeth mesh to give a **positive drive**. Advantages: exact ratio, more power, compact, high efficiency (≈ 98%). Disadvantages: costly, precise manufacture, need lubrication, unsuitable for distant shafts.

Types by shaft arrangement: parallel shafts — **spur** (teeth parallel to the axis, simple, noisy), **helical** (inclined teeth, gradual contact, quiet, but axial thrust), **herringbone/double helical** (axial thrust cancels), **rack and pinion** (rotary to linear motion). **Intersecting shafts** — **bevel** (conical, usually at 90°; equal size = **mitre**). **Neither parallel nor intersecting** — **worm and wheel** (very high ratio, self-locking), **spiral**, **hypoid** (car differential).

6.2 Gear terminology and the law of gearing

Pitch circle — an imaginary circle on which two gears are assumed to roll without slip; its diameter = **pitch diameter (D)**. **Circular pitch $p_c = \pi D/T$** (distance between corresponding points of adjacent teeth on the pitch circle; T = number of teeth). **Module m = D/T** (mm) — in SI this is what describes tooth size; $p_c = \pi m$. **Diametral pitch $P_d = T/D$** = 1/m. **Addendum** — from pitch circle to the top of the tooth (= 1m standard); **dedendum** — from pitch circle to the root (= 1.25m). **Clearance** = dedendum − addendum.

Fundamental law of gearing: for the angular velocity ratio of two gears to remain constant, the **common normal** to the tooth surfaces at the point of contact must always pass through a **fixed point — the pitch point** — on the line joining the centres. Tooth shapes satisfying this are called **conjugate profiles**. Two main profiles: **involute** and **cycloidal**.

Involute vs cycloidal: involute — the curve traced by the end of a string unwound from a circle (base circle); **constant pressure angle** (standard 20°), velocity ratio unchanged even if the centre distance varies slightly (a big advantage), made with one cutter, cheap — almost all gears today are involute. Cycloidal — pressure angle varies, less wear, stronger, but needs exact centre distance — used in clocks and instruments. **Interference** — in a pinion with very few teeth the tip of the gear tooth digs

into the pinion root; prevention — minimum number of teeth (≈ 18 for 20° full depth), increasing pressure angle.

Example: A gear has 40 teeth, module 5 mm. Pitch diameter, circular pitch, addendum and outside diameter?

Solution:

1. $D = mT = 5 \times 40 =$ **200 mm** ; $p_c = \pi m =$ **15.71 mm**
2. Addendum = m = 5 mm ; outside diameter = D + 2m = **210 mm**

6.3 Gear trains — how to find the speed ratio

For two meshing gears the pitch-line velocity is the same: $\pi D_1 N_1 = \pi D_2 N_2 \rightarrow$ $\mathbf{N_1/N_2 = D_2/D_1 = T_2/T_1}$ — the small gear (pinion) runs faster. External mesh reverses direction, internal mesh keeps it the same.

Derivation: Types of gear train

1. **Simple train:** one gear on each shaft ; $\mathbf{N_{first}/N_{last} = T_{last}/T_{first}}$ — intermediate gears (**idlers**) do not affect the ratio, only the direction (odd number of idlers → same direction)
2. **Compound train:** two gears on the same shaft rotate together ; **speed ratio = (product of teeth on driven gears)/(product of teeth on drivers)** — large ratio in small space
3. **Reverted train:** first and last gears on the same axis — clock hands, lathe back-gear ; condition: $r_1 + r_2 = r_3 + r_4$
4. **Epicyclic (planetary) train:** the axis of a gear itself revolves round another gear (sun, planet, arm, ring) — very high ratio, car automatic gearbox, differential ; analysed by the tabular method

Example: Compound train: driver 20 teeth → 60 teeth (on shaft 2) ; on the same shaft 25 teeth → 75 teeth (last). If the driver runs at 1200 rpm, final speed?

Solution:

1. Ratio = (60 × 75)/(20 × 25) = 4500/500 = 9
2. $N_{last} = 1200/9 =$ **133.3 rpm** ; two external meshes → same direction

1. Key Points

- **Classification by shaft position:** parallel shafts – spur, helical, herringbone; intersecting shafts – bevel (mitre gears when equal teeth at 90°); non-parallel, non-intersecting shafts – worm and worm wheel, spiral (crossed helical) gears.
- **Module:** m = D/T (mm). **Circular pitch:** p = πD/T = πm. **Diametral pitch:** P_d = T/D; hence p × P_d = π.
- **20° full-depth system:** addendum = 1m, dedendum = 1.25m, clearance = 0.25m, working depth = 2m, whole depth = 2.25m.
- **Law of gearing:** the common normal at the point of contact must always pass through the fixed pitch point, for a constant velocity ratio.

- **Involute profile:** the normal to an involute is tangent to the base circle; pressure angle remains constant and a small change in centre distance does not change the velocity ratio.
- **Interference:** occurs when the tip of one tooth digs into the flank of the mating tooth below the base circle. Minimum teeth to avoid interference: 32 for 14½°, 18 for 20° full depth, 14 for 20° stub teeth.
- **Contact ratio:** arc of contact ÷ circular pitch; it must be greater than 1 for continuous contact (usually 1.4–1.8).
- **Gear trains:** simple, compound, reverted (input and output shafts coaxial) and epicyclic (used in differentials, automatic transmissions).
- **Helical gears:** smooth and quiet but produce axial thrust; herringbone (double helical) gears cancel the axial thrust.
- **Pitch line velocity:** low velocity < 3 m/s, medium 3–15 m/s, high > 15 m/s.

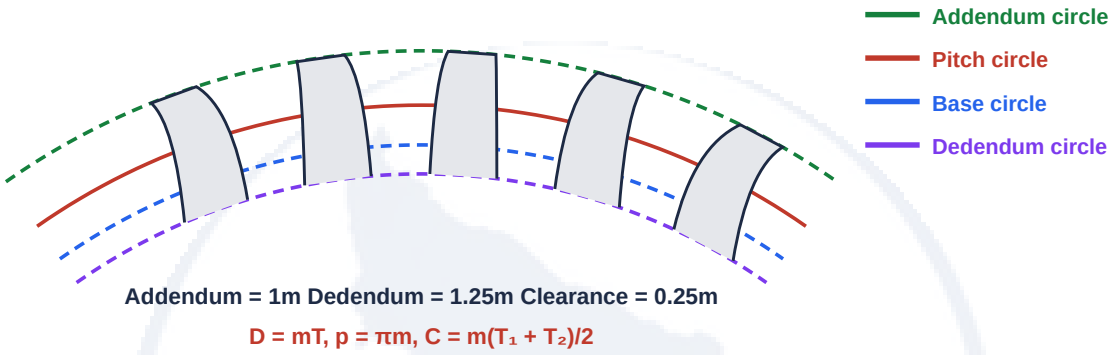


Fig 6.1: Gear tooth terminology (20° full-depth system)

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Gear ratio	**$G = T_g/T_p = N_p/N_g$**
Pitch circle diameter	**$D = m\,T$**
Circular pitch	**$p = \pi\,D / T = \pi\,m$**
Centre distance	**$C = m(T_1 + T_2)/2$**
Addendum circle diameter	**$D_a = m(T + 2)$**
Contact ratio	**Arc of contact / circular pitch (> 1)**
Maximum efficiency of spiral gears	**$[\cos(\theta+\varphi) + 1] / [\cos(\theta-\varphi) + 1]$**

3. Detailed Explanation

- **Path and arc of contact:** path of contact = path of approach + path of recess; arc of contact = path of contact / cos φ.

- **Simple gear train:** intermediate (idler) gears do not change the speed ratio, only the direction. With an odd number of gears the first and last rotate in the same direction; with an even number, in opposite directions.
- **Compound train:** speed ratio = product of driven teeth ÷ product of driver teeth. **Reverted train:** for the same module, $T_1 + T_2 = T_3 + T_4$.
- **Epicyclic train:** the axis of at least one gear moves; a large speed ratio is possible in a small space. The automobile differential is an epicyclic train.
- **Strength:** when gear and pinion are of the same material the pinion is weaker (fewer teeth, more load cycles), so the load capacity is decided by the pinion (Lewis equation).
- **Worm gear:** very high speed reduction in one stage, usually self-locking, low efficiency; used where shafts are non-parallel and non-intersecting.
- **Cycloidal profile:** no interference, but needs exact centre distance; used in clocks and instruments.

4. Comparison Table — Involute vs Cycloidal Profile

Basis	Involute	Cycloidal
Pressure angle	Constant	Variable
Change of centre distance	No effect on velocity ratio	Exact distance needed
Interference	May occur	Does not occur
Manufacture	Easy (single curve, rack cutter)	Difficult (two curves)
Use	General machines (most common)	Clocks, instruments

5. Solved Numerical Problems

Example 1. Module 5 mm, pinion 20 teeth and gear ratio 2.5. Find gear teeth, both pitch circle diameters, centre distance, addendum circle diameter of pinion and circular pitch.

Solution:

1. $T_g = 2.5 \times 20 =$ **50**; $D_p = 5 \times 20 =$ **100 mm**, $D_g = 5 \times 50 =$ **250 mm**
2. $C = (100 + 250)/2 =$ **175 mm**
3. Pinion $D_a = m(T + 2) = 5 \times 22 =$ **110 mm**; $p = \pi m =$ **15.71 mm**

Example 2. In a compound gear train, gear A (20 teeth) drives gear B (60 teeth). Gear C (25 teeth) on B's shaft drives gear D (75 teeth). A runs at 1800 rpm. Find the speed of D.

Solution:

1. Speed ratio $= (T_B \times T_D)/(T_A \times T_C) = (60 \times 75)/(20 \times 25) =$ **9**
2. $N_D = 1800/9 =$ **200 rpm**

Example 3. In a reverted gear train (same module) $T_1 = 20$, $T_2 = 40$ and $T_3 = 25$. Find T_4 and the overall speed ratio.

Solution:

1. Coaxial condition: $T_1 + T_2 = T_3 + T_4$, so $T_4 = 60 - 25 = \mathbf{35}$
2. Speed ratio = $(40 \times 35)/(20 \times 25) = \mathbf{2.8}$

Example 4. A simple gear train has 5 gears. The first gear rotates clockwise. State the direction of the last gear.

Solution:

1. The number of gears is odd, so the first and last gears rotate in the **same direction** (clockwise)
2. With an even number (e.g. 12 gears) the directions are opposite (UPSSSC JE 2016)

6. Frequently Asked Facts

- Module $m = D/T$; circular pitch $p = \pi m$; $p \times P_d = \pi$.
- Minimum 18 teeth for 20° full-depth involute to avoid interference.
- Normal to an involute is tangent to the base circle.

7. Note (Confusing Facts)

- Idler gears change the direction only, not the speed ratio.
- Herringbone gears cancel axial thrust; single helical gears produce it.
- Contact ratio must be **greater** than 1, not equal to 1.

Remember: One line: worm gear – non-parallel, non-intersecting shafts with high reduction. Assertion–Reason: (A) Involute gears are preferred in general machines. (R) A small change in centre distance does not change their velocity ratio. — Both true; R explains A.

30 Most Important One-Liner Facts — Gears, Types, Profiles & Gear Ratio

1. Formula of module?

— $m = D/T$.

2. Circular pitch?

— $p = \pi m$.

3. Diametral pitch?

— $P_d = T/D$.

4. Product of circular and diametral pitch?

— π.

5. Addendum in 20° full-depth system?

— 1m.

6. Dedendum in 20° full-depth system?

— 1.25m.

7. Clearance?

— 0.25m.

8. Gears for parallel shafts?

— Spur, helical, herringbone.

9. Gears for intersecting shafts?

— Bevel gears.

10. Bevel gears with equal teeth at 90°?

— Mitre gears.

11. Gears for non-parallel, non-intersecting shafts?

— Worm and worm wheel, spiral gears.

12. Gear that cancels axial thrust?

— Herringbone (double helical).

13. Law of gearing?

— Common normal always passes through the pitch point.

14. Normal to an involute is tangent to?

— Base circle.

15. Pressure angle of involute teeth?

— Constant.

16. Profile used in clocks?

— Cycloidal.

17. Minimum teeth to avoid interference (20° FD)?

— 18.

18. Minimum teeth to avoid interference (14½°)?

— 32.

19. Contact ratio?

— Arc of contact ÷ circular pitch.

20. Arc of contact?

— Path of contact ÷ cos φ.

21. Gear ratio?

— T_g/T_p.

22. Centre distance?

— $m(T_1 + T_2)/2$.

23. Effect of idler gear?

— Changes direction only.

24. Odd number of gears in a simple train?

— First and last rotate in same direction.

25. Condition of a reverted train?

— $T_1 + T_2 = T_3 + T_4$.

26. Gear train in an automobile differential?

— Epicyclic.

27. Gear that decides load capacity (same material)?

— Pinion.

28. Medium velocity gears range?

— 3–15 m/s.

29. Point common to both pitch circles?

— Pitch point.

30. Speed ratio of a gear train?

— Speed of driver ÷ speed of driven.

Practice MCQs — Gears, Types, Profiles & Gear Ratio

Q1. The formula of module is:

(a) T/D
(b) D/T
(c) πD/T
(d) πT/D

✔ **Answer: (b) D/T**

Explanation: Module = pitch circle diameter ÷ number of teeth.

Q2. Addendum in the 20° full-depth system is:

(a) 1m
(b) 1.25m
(c) 0.25m
(d) 2.25m

✔ **Answer: (a) 1m**

Explanation: In the 20° full-depth system, addendum = 1 module.

Q3. The clearance is:

(a) 0.157m
(b) 0.25m
(c) 1m
(d) 1.25m

✔ **Answer: (b) 0.25m**

Explanation: Clearance = dedendum − addendum = 1.25m − 1m = 0.25m.

Q4. The normal to an involute is tangent to the:

(a) Pitch circle
(b) Base circle
(c) Addendum circle
(d) Root circle

✔ **Answer: (b) Base circle**

Explanation: An involute is generated by a line rolling on the base circle, so its normal is tangent to the base circle.

Q5. Mitre gears are:

(a) Spur
(b) Bevel (equal teeth, 90°)
(c) Worm
(d) Helical

✔ **Answer: (b) Bevel (equal teeth, 90°)**

Explanation: Mitre gears are bevel gears of equal size with shafts at 90°.

Q6. Axial thrust is cancelled in:

(a) Spur
(b) Helical
(c) Herringbone
(d) Bevel

✔ **Answer: (c) Herringbone**

Explanation: The two opposite helices of a herringbone gear cancel each other's axial thrust.

Q7. Profile used in clocks:

(a) Involute
(b) Cycloidal
(c) Stub
(d) Straight

✔ **Answer: (b) Cycloidal**

Explanation: Cycloidal teeth have no interference and are used in clocks.

Q8. Gear train used in an automobile differential:

(a) Simple
(b) Compound
(c) Reverted
(d) Epicyclic

✔ **Answer: (d) Epicyclic**

Explanation: The differential uses an epicyclic train to allow different wheel speeds.

Q9. Circular pitch × diametral pitch =

(a) 1
(b) π
(c) 2π
(d) π/2

✔ **Answer: (b) π**

Explanation: $p \times P_d = (\pi D/T) \times (T/D) = \pi$.

Q10. Minimum teeth to avoid interference (20° full depth):

(a) 12 (b) 14

(c) 18 (d) 32

✔ **Answer: (c) 18**

Explanation: For 20° full-depth involute teeth, at least 18 teeth on the pinion avoid interference.

SSC JE (Mechanical) Previous Year Questions — Gears, Types, Profiles & Gear Ratio

Q1. The point common to both pitch circles is called:

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(a) Pitch point (b) Base point

(c) Contact point (d) Involute point

✔ **Answer: (a) Pitch point**

Explanation: The point where the two pitch circles touch is the pitch point.

Q2. For 20° involute gears, path of approach 28 mm and recess 25 mm. Arc of contact (mm) is (cos 20° = 0.94):

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(a) 54 (b) 54.4

(c) 56 (d) 56.4

✔ **Answer: (d) 56.4**

Explanation: Arc of contact = (28 + 25)/cos 20° = 53/0.94 = 56.4 mm.

Q3. Contact ratio of gears with module 12 mm and arc of contact 60 mm:

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(a) 2 (b) 1.2

(c) 1.6 (d) 5

✔ **Answer: (c) 1.6**

Explanation: Contact ratio = arc of contact/(πm) = 60/(π × 12) = 1.6.

Q4. Correct statement about the length of arc of contact:

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(a) Directly proportional to path of contact (b) Independent of path of contact

(c) Inversely proportional to path of contact (d) Always equal to path of contact

✔ **Answer: (a) Directly proportional to path of contact**

Explanation: Arc of contact = path of contact/cos φ, so it is directly proportional to the path of contact.

Q5. Speed ratio of a gear train is:

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(a) Speed of driver ÷ speed of driven (b) Speed of driven ÷ speed of driver

(c) Teeth on driver ÷ teeth on driven (d) None of these

✔ **Answer: (a) Speed of driver ÷ speed of driven**

Explanation: Speed ratio = speed of driver ÷ speed of driven.

Chapter 7: Governors (Principle & Classification)

Governors (Principle & Classification) – Basic Concept

A **governor** automatically controls the supply of working fluid (fuel or steam) to an engine so that its **mean speed** remains within limits when the load changes. When the load decreases the speed rises, the balls fly outward, the sleeve moves and the fuel supply is reduced; the reverse happens when the load increases.

Understand in Detail

7.1 Governor — constant speed even when the load changes

Think of a diesel generator supplying a village. In the evening as all lights come on the load rises — if the fuel stays the same the engine will slow down and the frequency will fall. At night as the load drops the engine will race. A **governor** senses the engine speed and automatically decreases or increases the **fuel/steam supply** so that the **mean speed** stays nearly constant even as the load changes.

Principle of the centrifugal governor: two heavy balls (fly-balls) hang by arms from a vertical spindle driven by the engine. Speed rises → centrifugal force rises → balls move out and up → the **sleeve** below slides up → through a lever the throttle valve closes → less fuel → speed falls. When speed falls, exactly the reverse. Thus **ball position = speed signal**.

Classification: centrifugal (most) and **inertia** — sensitive to acceleration, quick response. Centrifugal: **loaded (pendulum/dead-weight)** — **Watt** (simplest, no load on sleeve), **Porter** (central load on sleeve), **Proell** (like Porter but balls on an extension of the lower arm); **spring-controlled** — **Hartnell**, **Hartung**, **Wilson-Hartnell**, **Pickering** — for high speeds, can be mounted in any position.

Derivation: Height of Watt and Porter governors

1. **Governor height h** = vertical distance from the plane of the balls to the point where the arms meet the axis
2. **Watt:** equilibrium of a ball (weight mg, centrifugal $m\omega^2 r$, arm tension) → $\tan\theta = r/h = \omega^2 r/g$ → $\mathbf{h = g/\omega^2 = 895/N^2\ m}$ (N in rpm)
3. Meaning: speed doubles → h becomes a quarter ; at high speed h becomes tiny and sensitivity is lost — so Watt is only for low speeds (≈ 60–80 rpm)
4. **Porter** (central load M, ball m, equal arms): $\mathbf{h = (g/\omega^2)(m + M)/m}$ → $\mathbf{N^2 = (895/h)(m + M)/m}$ — adding M gives a higher speed for the same h, so it suits higher speeds

7.2 Governor characteristics — sensitiveness, stability, isochronism, hunting

Sensitiveness — a large sleeve movement for a small change in speed; measure: $(N_1 - N_2)/N$ (the smaller, the more sensitive). **Stability** — a definite ball position for every speed; as speed increases the

radius also increases. **Isochronous governor** — one with the same equilibrium speed over its whole working range (range zero) — i.e. **infinitely sensitive**, but in practice it **hunts**.

Hunting — an over-sensitive governor fully opens or fully closes the fuel for a tiny speed change; the engine speed keeps swinging above and below the required value. Remedy — a less sensitive governor or a dashpot (damper). **Effort** — mean force on the sleeve for a 1% change in speed; **power** — effort × sleeve displacement. **Controlling force** — the force pulling the ball inward; its curve is used to check stability.

> **Example:** A Watt governor runs at 60 rpm. Height? If speed rises to 61 rpm, how much does the sleeve rise (approx. 2 × change in height)?
> **Solution:**
> 1. $h_1 = 895/60^2 =$ **0.2486 m** ; $h_2 = 895/61^2 = 0.2405$ m
> 2. $\Delta h = 8.1$ mm $\rightarrow$ sleeve lift $\approx 2\Delta h =$ **16.2 mm**

1. Key Points

- **Classification:** centrifugal governors – (a) pendulum type: Watt; (b) loaded type: dead-weight loaded (Porter, Proell) and spring loaded (Hartnell, Hartung, Wilson-Hartnell, Pickering). Inertia governors form a separate class.
- **Watt governor:** simplest pendulum type; height $h = g/\omega^2 = 895/N^2$ m. Suitable only for low speeds (the height becomes very small at high speed).
- **Porter governor:** a Watt governor with a heavy central load M on the sleeve; $h = [(m + M)/m] \times 895/N^2$ (equal arms). It can run at higher speeds.
- **Proell governor:** balls fixed on an extension of the lower arms; more sensitive than a Porter governor of the same size.
- **Hartnell governor:** spring-loaded with bell-crank levers; can be mounted in any position and run at high speeds.
- **Sensitiveness:** $(N_2 - N_1)/N$; a governor is sensitive if a small change of speed produces a large sleeve movement.
- **Stability:** a governor is stable if the radius of rotation increases as the equilibrium speed increases. If the radius decreases with increasing speed, it is unstable.
- **Isochronism:** the equilibrium speed is the same for all radii (range of speed zero). An isochronous governor is infinitely sensitive and leads to **hunting**. Only spring-loaded governors (e.g. Hartnell) can be isochronous.
- **Hunting:** continuous fluctuation of engine speed above and below the mean, caused by a governor that is too sensitive.
- **Effort and power:** effort = mean force on the sleeve for a given fractional change of speed; power = effort × lift of sleeve.

- **Controlling force:** F = mω²r; for stability the controlling-force curve (F vs r) must have the form F = ar − b; F = ar + b gives an unstable governor and F = ar an isochronous one.
- **Insensitiveness:** due to friction, there is a range of speed (N″ − N′) over which the sleeve does not move; coefficient of insensitiveness = (N″ − N′)/N.

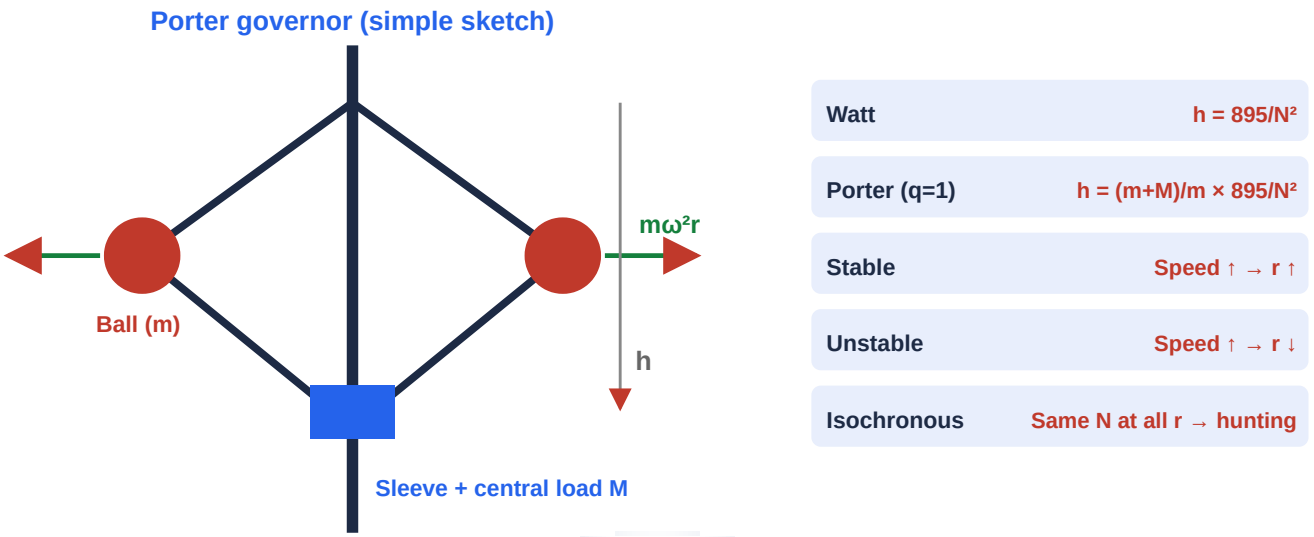


Fig 7.1: Porter governor and key formulas

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Watt governor	**$h = g/\omega^2 = 895/N^2$ (m)**
Porter governor (q = 1)	**$h = [(m + M)/m] \times 895/N^2$**
Sensitiveness	**$(N_2 - N_1)/N$**
Controlling force	**$F = m\omega^2 r$**
Coefficient of insensitiveness	**(N'' − N')/N**

3. Detailed Explanation

- **Working principle:** centrifugal force on the rotating balls is balanced by gravity (Watt, Porter) or spring force (Hartnell). Any change of speed changes this balance, moving the sleeve, which operates the throttle through levers.
- **Why Watt is limited to low speeds:** at 60 rpm h ≈ 0.25 m, but at 120 rpm h ≈ 0.06 m; the change in height for a given change in speed becomes too small to operate the throttle.
- **Hartnell governor stiffness:** s = $(F_2 - F_1)$/lift (with equal arms of the bell-crank lever). Increasing the stiffness makes it more stable but less sensitive.
- **Converting an unstable spring-controlled governor:** increase the spring stiffness (or change initial compression) so that the controlling-force line takes the form F = ar − b.
- **Governor vs flywheel:** the governor regulates the mean speed over many cycles by controlling the fuel; the flywheel only smooths the speed within each cycle and has no control on fuel.

- **Sensitive vs stable:** these are opposing requirements; a governor that is too sensitive tends to hunt, so a compromise is chosen.

4. Comparison Table — Comparison of Main Governors

Governor	Type	Feature
Watt	Pendulum (centrifugal)	$h = 895/N^2$; low speeds only
Porter	Dead-weight	Central load M; $h = (m+M)/m \times 895/N^2$
Proell	Dead-weight	Balls on extension of lower arm; more sensitive than Porter
Hartnell	Spring loaded	Bell-crank lever; high speed, any position
Pickering	Spring loaded	Small machines (gramophone)

5. Solved Numerical Problems

Example 1. Find the height of a Watt governor at 60 rpm and 120 rpm.
Solution:

1. At 60 rpm: $h = 895/60^2 = 895/3600 =$ **0.249 m (about 24.9 cm)**
2. At 120 rpm: $h = 895/14400 =$ **0.062 m (about 6.2 cm)**; doubling the speed reduces the height to one-fourth

Example 2. A Porter governor (equal arms) has m = 4 kg, M = 20 kg and h = 0.2 m. Find the equilibrium speed.
Solution:

1. $N^2 = [(m + M)/m] \times 895/h = 6 \times 895/0.2 = 26850$
2. $N = \sqrt{26850} =$ **about 163.9 rpm**

Example 3. The minimum and maximum equilibrium speeds of a governor are 290 rpm and 310 rpm. Find the sensitiveness.
Solution:

1. $N = (290 + 310)/2 = 300$ rpm
2. Sensitiveness $= (310 - 290)/300 =$ **0.0667 (6.67%)**

Example 4. A ball of mass 2 kg rotates at radius 0.15 m at 300 rpm. Find the controlling force.
Solution:

1. $\omega = 31.42$ rad/s, so $\omega^2 = 986.96$
2. $F = m\omega^2 r = 2 \times 986.96 \times 0.15 =$ **about 296 N**

6. Frequently Asked Facts

- Watt: $h = 895/N^2$ m – height $\propto 1/N^2$.
- Isochronous governor → hunting; only spring-loaded governors can be isochronous.

- Friction causes insensitiveness; increasing spring stiffness makes a governor less sensitive.

7. Note (Confusing Facts)

- Unstable: radius decreases as speed increases. Stable: radius increases as speed increases.
- Sensitiveness and stability oppose each other – too sensitive means hunting.
- A governor controls the **mean** speed; it does not remove fluctuation within a cycle.

Remember: One line: Hartnell – spring loaded, high speed, can be mounted in any position. Assertion–Reason: (A) Watt governor is unsuitable for high speeds. (R) Its height varies as $1/N^2$, becoming very small at high speed. — Both true; R explains A.

30 Most Important One-Liner Facts — Governors (Principle & Classification)

1. Function of a governor?

— To control the mean speed of an engine.

2. Type of Watt governor?

— Pendulum (centrifugal) type.

3. Height of a Watt governor?

— $h = 895/N^2$ m.

4. Height of a Watt governor is proportional to?

— $1/N^2$.

5. Watt governor suitable for?

— Low speeds only.

6. Porter governor is which type?

— Dead-weight loaded.

7. Height of Porter governor?

— $[(m + M)/m] \times 895/N^2$.

8. Ratio of Porter to Watt height?

— $(m + M)/m$.

9. Governor with balls on extension of lower arm?

— Proell governor.

10. Proell compared with Porter?

— More sensitive.

11. Hartnell governor is which type?

— Spring loaded.

12. Governor mountable in any position?

— Hartnell.

13. Governor for gramophones?

— Pickering governor.

14. Sensitiveness?

— $(N_2 - N_1)/N$.

15. Condition of stability?

— Radius increases with speed.

16. Condition of an unstable governor?

— Radius decreases with speed.

17. Isochronous governor?

— Same equilibrium speed at all radii.

18. Isochronous governor leads to?

— Hunting.

19. Governor that can be isochronous?

— Spring loaded (Hartnell).

20. Hunting?

— Continuous speed fluctuation above and below mean.

21. Cause of insensitiveness?

— Friction.

22. Coefficient of insensitiveness?

— $(N'' - N')/N$.

23. Power of a governor?

— Effort × lift of sleeve.

24. Controlling force?

— $F = m\omega^2 r$.

25. F = ar + b gives which governor?

— Unstable.

26. F = ar gives which governor?

— Isochronous.

27. Effect of increasing spring stiffness?

— More stable, less sensitive.

28. When speed rises, balls move?

— Outward (radius increases).

29. When speed rises, height of Watt governor?

— Decreases.

30. Governor vs flywheel – who controls fuel?

— Governor.

Practice MCQs — Governors (Principle & Classification)

Q1. The Watt governor is of which type?

(a) Spring loaded
(b) Pendulum type
(c) Inertia
(d) Dead weight

✔ **Answer: (b) Pendulum type**

Explanation: The Watt governor is the simple pendulum (centrifugal) type.

Q2. The height of a Watt governor is proportional to:

(a) N
(b) N^2
(c) 1/N
(d) $1/N^2$

✔ **Answer: (d) $1/N^2$**

Explanation: $h = g/\omega^2$, so $h \propto 1/N^2$.

Q3. The Hartnell governor is:

(a) Dead weight
(b) Spring loaded
(c) Pendulum
(d) Inertia

✔ **Answer: (b) Spring loaded**

Explanation: Hartnell uses a spring to provide the controlling force.

Q4. If the radius decreases as speed increases, the governor is:

(a) Stable
(b) Unstable
(c) Isochronous
(d) Sensitive

✔ **Answer: (b) Unstable**

Explanation: If radius falls as speed rises, the governor is unstable.

Q5. An isochronous governor results in:

(a) Stability
(b) Hunting
(c) Insensitiveness
(d) High effort

✔ **Answer: (b) Hunting**

Explanation: Infinite sensitivity of an isochronous governor makes the engine hunt.

Q6. Height of a Porter governor compared to a Watt governor:

(a) Less
(b) (m + M)/m times
(c) Same
(d) Half

✔ **Answer: (b) (m + M)/m times**

Explanation: The central load M raises the height by the factor (m + M)/m.

Q7. Power of a governor =

(a) Effort × lift
(b) Effort ÷ lift
(c) $m\omega^2 r$
(d) Effort × speed

✔ **Answer: (a) Effort × lift**

Explanation: Power = mean effort × lift of the sleeve.

Q8. Governor that can be mounted in any position:

(a) Watt
(b) Porter
(c) Hartnell
(d) Proell

✔ **Answer: (c) Hartnell**

Explanation: Spring loading lets the Hartnell governor work in any position.

Q9. Friction in a governor causes:

(a) Hunting
(b) Insensitiveness
(c) Isochronism
(d) Stability

✔ **Answer: (b) Insensitiveness**

Explanation: Friction creates a range of speed with no sleeve movement – insensitiveness.

Q10. Compared to Porter, the Proell governor is:

(a) Less sensitive
(b) More sensitive
(c) Same
(d) Isochronous

✔ **Answer: (b) More sensitive**

Explanation: Balls on the extension of the lower arms make the Proell more sensitive.

SSC JE (Mechanical) Previous Year Questions — Governors (Principle & Classification)

Q1. Which statement is correct regarding a governor?

SSC JE 2023 (9 October, Shift-2)

(a) It controls acceleration of the prime mover
(b) It controls temperature of the prime mover
(c) It controls mean speed of the prime mover
(d) It is not a compulsory device

✔ **Answer: (c) It controls mean speed of the prime mover**

Explanation: A governor regulates the mean speed by controlling the fuel supply.

Q2. Method to convert a spring-controlled governor from unstable to stable:

SSC JE 2018 (23 January, Morning)

(a) Increasing ball weight
(b) Decreasing spring stiffness
(c) Decreasing ball weight
(d) Increasing spring stiffness

✔ **Answer: (d) Increasing spring stiffness**

Explanation: Greater spring stiffness changes F = ar + b into a stable characteristic.

Q3. Governor with large sleeve displacement for a small change of speed is:

SSC JE 2018 (23 January, Morning)

(a) Sensitive
(b) Stable
(c) Isochronous
(d) Unstable

✔ **Answer: (a) Sensitive**

Explanation: Large sleeve movement for a small speed change means the governor is sensitive.

Q4. Which is NOT a loaded-type governor?

SSC JE 2018 (23 January, Evening)

(a) Dead weight governor
(b) Spring-controlled governor
(c) Watt governor
(d) Porter governor

✔ **Answer: (c) Watt governor**

Explanation: Watt is a simple pendulum type with no central load or spring.

Q5. If Fc = a·r + b for a spring-controlled governor, it is:

SSC JE 2018 (22 January, Evening)

(a) Stable
(b) Unstable
(c) Isochronous
(d) Sensitive

✔ **Answer: (b) Unstable**

Explanation: Positive intercept b in F = ar + b gives an unstable governor.

Q6. Height of a Watt governor in decametres:

SSC JE 2018 (22 January, Evening)

(a) $8.95/N^2$
(b) $89.5/N^2$
(c) $895/N^2$
(d) $8950/N^2$

✔ **Answer: (b) $89.5/N^2$**

Explanation: $h = 895/N^2$ m = $89.5/N^2$ decametres.

Q7. Ratio of height of Porter to Watt governor (equal arms):

SSC JE 2018 (22 January, Evening)

(a) (M + m)/m
(b) m/(M + m)
(c) M/m
(d) (M − m)/m

✔ **Answer: (a) (M + m)/m**

Explanation: For equal arms, h_{Porter}/h_{Watt} = (M + m)/m.

Q8. Which is a gravity-controlled governor? SSC JE (Previous Year)

(a) Hartnell
(b) Hartung
(c) Watt
(d) Pickering

✔ **Answer: (c) Watt**

Explanation: The Watt governor works by gravity (no spring).

Q9. Hartnell governor: 800 N at min radius, 1200 N at max radius, sleeve lift 20 mm. Spring stiffness (N/mm): SSC JE (Previous Year)

(a) 10
(b) 20
(c) 15
(d) 18

✔ **Answer: (b) 20**

Explanation: Stiffness = (1200 − 800)/20 = 20 N/mm.

Q10. Angular speed (rad/s) of a Watt governor of height 20 cm: SSC JE (Previous Year)

(a) 20
(b) 10
(c) 6
(d) 7

✔ **Answer: (d) 7**

Explanation: $\omega^2 = g/h = 9.81/0.2 = 49$, so $\omega = 7$ rad/s.

Q11. When the speed of a governor increases: SSC JE (Previous Year)

(a) Height and radius both increase
(b) Height and radius both decrease
(c) Height decreases, radius increases
(d) Height increases, radius decreases

✔ **Answer: (c) Height decreases, radius increases**

Explanation: Higher speed → balls fly out (radius ↑) and height ↓.

Q12. Which governor can be isochronous? SSC JE (Previous Year)

(a) Watt
(b) Porter
(c) Pickering
(d) Hartnell

✔ **Answer: (d) Hartnell**

Explanation: Only a spring-loaded governor such as the Hartnell can be made isochronous.

Chapter 8: Cams & Followers

Cams & Followers – Basic Concept

A **cam** is a rotating machine element that gives a reciprocating or oscillating motion to another element called the **follower**, by direct contact. Cams are used to operate the inlet and exhaust valves of I.C. engines, in automatic machines, printing machines and textile machinery.

Understand in Detail

8.1 Cam and follower — any desired motion from rotation

In an engine, how is a valve opened exactly at the right time, kept open for a while and then closed? By a **cam**. A cam is a specially shaped rotating part that gives the **follower** in contact with it a reciprocating or oscillating motion according to a definite law. Cam-follower is a **higher pair** (line/point contact), kept in contact by a spring (**force-closed**). Uses: IC engine valve gear, automatic machines, sewing machines, textile mills.

Types of cams: radial/disc — follower moves perpendicular to the axis (most common); **cylindrical** — follower moves parallel to the axis; **tangent, circular-arc, inverted cam**. **Types of followers:** by surface — **knife edge** (rapid wear, rare), **roller** (less wear, most used), **flat/mushroom** (engine valves — compact), **spherical**; by motion — translating and oscillating; by axis — **radial** (follower line passes through cam centre) and **offset**.

8.2 Cam terminology and laws of follower motion

Base circle — the smallest circle touching the cam profile. **Trace point** — the point of the follower used to generate the pitch curve (roller centre). **Prime circle** — the smallest circle touching the pitch curve; for a roller follower = base circle radius + roller radius. **Pressure angle** — the angle between the direction of follower motion and the normal at the contact point; it should **not exceed 30°**, otherwise the follower jams in its guide. **Lift/stroke** — maximum displacement of the follower. **Rise (outstroke), dwell, return** — parts of one cam revolution.

Derivation: Four standard laws of follower motion

1. **Uniform velocity:** displacement diagram a straight line ; but at the start and end of rise the velocity changes suddenly → **infinite acceleration** (shock) — so used with rounded ends (modified)
2. **Simple harmonic motion (SHM):** displacement $s = (h/2)(1 - \cos(\pi\theta/\theta_0))$; **maximum velocity = $\pi h\omega/(2\theta_0)$** ; **maximum acceleration = $\pi^2 h\omega^2/(2\theta_0^2)$** — acceleration at the ends is finite but changes suddenly (jerk)
3. **Uniform acceleration and retardation (parabolic):** constant acceleration for half the rise, constant retardation for the other half ; maximum velocity = $2h\omega/\theta_0$; acceleration = $4h\omega^2/\theta_0^2$ — **the least acceleration for a given lift**
4. **Cycloidal:** $s = h[\theta/\theta_0 - \sin(2\pi\theta/\theta_0)/(2\pi)]$; acceleration at the ends is **zero** → no shock — **best for high-speed cams** ; maximum velocity = $2h\omega/\theta_0$, maximum acceleration = $2\pi h\omega^2/\theta_0^2$

Example: A cam rotates at 300 rpm; the follower rises 40 mm with SHM during 120° of cam rotation. Maximum velocity and acceleration?

Solution:

1. $\omega = 2\pi \times 300/60 = 31.42$ rad/s ; $\theta_0 = 120° = 2.094$ rad ; $h = 0.04$ m
2. $v_{max} = \pi \times 0.04 \times 31.42/(2 \times 2.094) =$ **0.943 m/s**
3. $a_{max} = \pi^2 \times 0.04 \times 987/(2 \times 4.386) =$ **44.4 m/s²**

1. Key Points

- **Cam–follower pair:** a higher pair (point/line contact), kept in contact by a spring – force-closed.
- **Types of followers by contact surface:** knife-edge (maximum wear), roller (most common, low friction), flat-faced/mushroom (automobile valves), spherical.
- **By motion:** reciprocating (translating) or oscillating; **by line of motion:** radial or offset (offset reduces side thrust).
- **Base circle:** the smallest circle drawn to the cam profile from the cam centre. **Prime circle** = base circle + roller radius. **Pitch curve:** the path traced by the trace point.
- **Lift (stroke/throw):** maximum displacement of the follower measured from the base circle.
- **Pressure angle:** the angle between the direction of follower motion and the normal to the pitch curve; it should be small (≤ 30°). Increasing the base circle reduces it.
- **Follower motions:** uniform velocity (infinite acceleration at ends – suitable only for slow speed), SHM, uniform acceleration and retardation (parabolic), cycloidal (no sudden change of acceleration – best for high speed).
- **Dwell:** the period during which the follower remains at rest while the cam rotates.

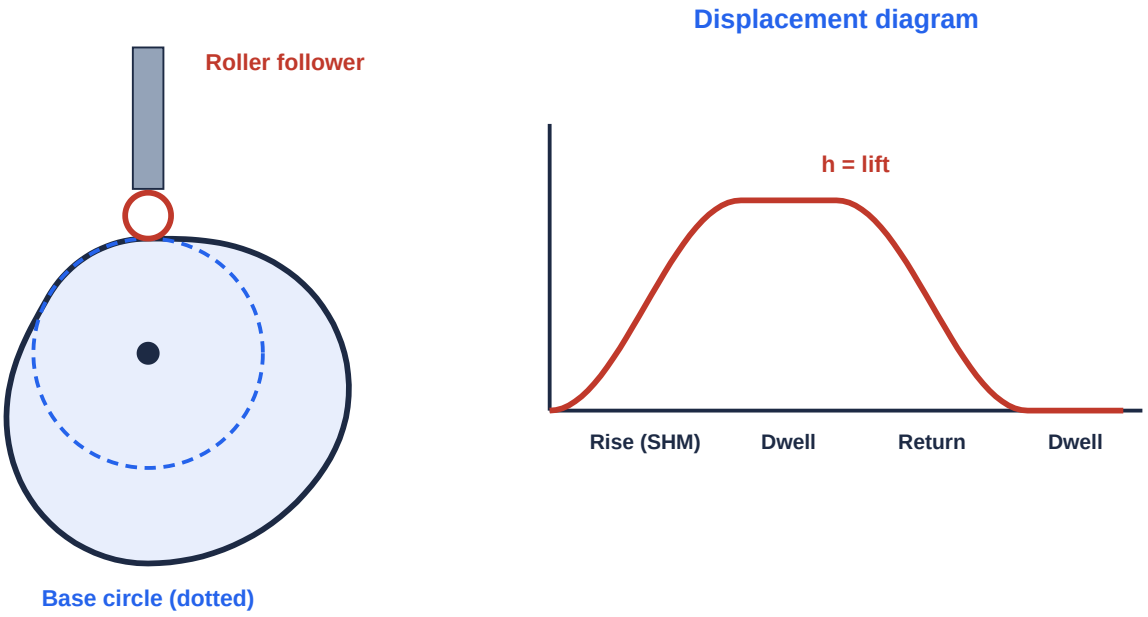


Fig 8.1: Disc cam, roller follower and displacement diagram

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
SHM: maximum velocity	$\pi h \omega / (2\theta_o)$
SHM: maximum acceleration	$\pi^2 h \omega^2 / (2\theta_o^2)$
Uniform acceleration: maximum velocity	$2 h \omega / \theta_o$
Uniform acceleration: acceleration	$4 h \omega^2 / \theta_o^2$
Cycloidal: maximum velocity	$2 h \omega / \theta_o$
Cycloidal: maximum acceleration	$2\pi h \omega^2 / \theta_o^2$

3. Detailed Explanation

- **Motion cycle:** a typical cam has four phases – rise (outstroke), dwell, return (instroke) and dwell. The displacement diagram plots follower displacement against cam angle.
- **Uniform velocity:** velocity changes suddenly at the start and end of the stroke, so the acceleration becomes theoretically infinite – causing shock. It is modified with circular arcs at the ends.
- **SHM:** maximum velocity at mid-stroke, maximum acceleration at the ends; the acceleration jumps at the start and end of the rise, giving finite jerk issues at high speed.
- **Cycloidal motion:** acceleration is zero at both ends and varies sinusoidally, so there is no sudden change – hence it is the best choice for high-speed cams (and an important PYQ).
- **Camshaft speed in a 4-stroke engine:** each valve opens once in two crank revolutions, so the camshaft runs at half the crankshaft speed.

4. Comparison Table — Comparison of Followers

Follower	Advantage	Drawback / Use
Knife-edge	Simple; works with any profile	Maximum wear; rarely used
Roller	Low friction and wear	Most widely used (gas and oil engines)
Flat-faced (mushroom)	Compact; low side thrust	Automobile engine valves
Spherical	Less wear than flat face	Automobile valves

5. Solved Numerical Problems

Example 1. The follower rises 30 mm with SHM during 120° of cam rotation. The cam runs at 300 rpm. Find the maximum velocity and acceleration of the follower.

Solution:

1. $\omega = 31.42$ rad/s, $\theta_o = 120° = 2.094$ rad
2. $v_{max} = \pi h\omega/(2\theta_o) = \pi \times 0.03 \times 31.42/(2 \times 2.094) =$ **0.707 m/s**
3. $a_{max} = \pi^2 h\omega^2/(2\theta_o^2) = 9.87 \times 0.03 \times 986.96/(2 \times 4.386) =$ **33.3 m/s²**

Example 2. For the same data, find the maximum velocity and acceleration with uniform acceleration and retardation.

Solution:

1. $v_{max} = 2h\omega/\theta_o = 2 \times 0.03 \times 31.42/2.094 =$ **0.900 m/s**
2. $a = 4h\omega^2/\theta_o^2 = 4 \times 0.03 \times 986.96/4.386 =$ **27.0 m/s²**

Example 3. For the same data, find the maximum velocity and acceleration with cycloidal motion.

Solution:

1. $v_{max} = 2h\omega/\theta_o =$ **0.900 m/s**
2. $a_{max} = 2\pi h\omega^2/\theta_o^2 = 6.283 \times 0.03 \times 986.96/4.386 =$ **42.4 m/s²**
3. Note: the peak acceleration of cycloidal motion is higher, but it is zero at the ends, so there is no shock

6. Frequently Asked Facts

- Cycloidal motion is best for high-speed cams.
- Throw/lift is measured from the base circle; prime circle = base circle + roller radius.
- Roller follower is the most widely used; knife-edge wears the most.

7. Note (Confusing Facts)

- Uniform velocity motion gives **infinite** acceleration at the ends, not zero.
- Offset of the follower reduces side thrust; increasing the base circle reduces pressure angle.
- Camshaft speed = half the crankshaft speed in a four-stroke engine.

Remember: One line: cam and follower – higher pair, force-closed by a spring. Assertion–Reason: (A) Cycloidal motion is preferred for high-speed cams. (R) Its acceleration is zero at the beginning and end of the stroke. — Both true; R explains A.

30 Most Important One-Liner Facts — Cams & Followers

1. Type of pair formed by cam and follower?

— Higher pair.

2. How is a follower kept in contact?

— By a spring (force-closed).

3. Follower with maximum wear?

— Knife-edge follower.

4. Most widely used follower?

— Roller follower.

5. Follower used for automobile valves?

— Flat-faced (mushroom).

6. Purpose of offset in a follower?

— To reduce side thrust.

7. Base circle?

— Smallest circle to the cam profile.

8. Prime circle?

— Base circle + roller radius.

9. Pitch curve?

— Path of the trace point.

10. Lift or throw is measured from?

— Base circle.

11. Pressure angle?

— Angle between follower motion and normal to pitch curve.

12. How to reduce pressure angle?

— Increase base circle.

13. Period when follower is at rest?

— Dwell.

14. Outward motion of follower?

— Rise (outstroke).

15. Acceleration at ends in uniform velocity motion?

— Infinite.

16. Uniform velocity motion is suitable for?

— Slow speeds only.

17. Maximum velocity in SHM occurs at?

— Mid-stroke.

18. Maximum acceleration in SHM occurs at?

— Ends of stroke.

19. SHM maximum velocity?

— $\pi h\omega/(2\theta_o)$.

20. SHM maximum acceleration?

— $\pi^2 h\omega^2/(2\theta_o^2)$.

21. Uniform acceleration – maximum velocity?

— $2h\omega/\theta_o$.

22. Uniform acceleration – acceleration?

— $4h\omega^2/\theta_o^2$.

23. Cycloidal maximum acceleration?

— $2\pi h\omega^2/\theta_o^2$.

24. Best motion for high-speed cams?

— Cycloidal.

25. Acceleration at ends in cycloidal motion?

— Zero.

26. Camshaft speed in a 4-stroke engine?

— Half the crankshaft speed.

27. Displacement diagram plots?

— Follower displacement vs cam angle.

28. Recommended maximum pressure angle?

— About 30°.

29. Applications of cams?

— Engine valves, automatic and textile machines.

30. Typical four phases of cam motion?

— Rise, dwell, return, dwell.

Practice MCQs — Cams & Followers

Q1. The cam-follower pair is a:

(a) Lower pair
(b) Higher pair
(c) Sliding pair
(d) Turning pair

✔ **Answer: (b) Higher pair**

Explanation: Cam and follower have point/line contact – a higher pair.

Q2. Follower with maximum wear:

(a) Roller
(b) Knife edge
(c) Flat
(d) Spherical

✔ **Answer: (b) Knife edge**

Explanation: The sharp knife edge has a very small contact area, so wear is maximum.

Q3. Best motion for high speed:

(a) Uniform velocity
(b) SHM
(c) Cycloidal
(d) Parabolic

✔ **Answer: (c) Cycloidal**

Explanation: Cycloidal motion has no sudden change of acceleration – best at high speed.

Q4. In uniform velocity motion, acceleration at the ends is:

(a) Zero
(b) Constant
(c) Infinite
(d) Negative

✔ **Answer: (c) Infinite**

Explanation: Velocity changes suddenly at the ends, so acceleration is theoretically infinite.

Q5. Prime circle =

(a) Base circle − roller radius
(b) Base circle + roller radius
(c) Pitch circle
(d) Base circle

✔ **Answer: (b) Base circle + roller radius**

Explanation: Prime circle radius = base circle radius + roller radius.

Q6. The throw of a cam is measured from the:

(a) Pitch circle
(b) Base circle
(c) Prime circle
(d) Pitch curve

✔ **Answer: (b) Base circle**

Explanation: Throw is the maximum follower displacement from the base circle.

Q7. Camshaft speed in a four-stroke engine is:

(a) Equal to crankshaft
(b) Half
(c) Double
(d) Four times

✔ **Answer: (b) Half**

Explanation: Each valve opens once in two crank revolutions, so camshaft speed = ½ crank speed.

Q8. The period when the follower is at rest:

(a) Rise
(b) Return
(c) Dwell
(d) Lift

✔ **Answer: (c) Dwell**

Explanation: When the follower does not move while the cam turns, it is called dwell.

Q9. Method to reduce the pressure angle:

(a) Increase base circle
(b) Increase lift
(c) Increase speed
(d) Use knife-edge follower

✔ **Answer: (a) Increase base circle**

Explanation: A larger base circle makes the profile flatter and reduces the pressure angle.

Q10. In SHM, the maximum velocity occurs:

(a) At start of stroke
(b) At mid-stroke
(c) At end of stroke
(d) During dwell

✔ **Answer: (b) At mid-stroke**

Explanation: In SHM, velocity is maximum at mid-stroke where displacement is half.

SSC JE (Mechanical) Previous Year Questions — Cams & Followers

Q1. Purpose of offset in a cam-follower mechanism:

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(a) To minimise side thrust
(b) To increase lift
(c) To increase speed
(d) To increase wear

✔ **Answer: (a) To minimise side thrust**

Explanation: Offsetting the follower axis reduces the side thrust on the follower guide.

Q2. Contact between cam and follower forms a:

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(a) Lower pair
(b) Higher pair
(c) Sliding pair
(d) Rolling pair

✔ **Answer: (b) Higher pair**

Explanation: Point or line contact between cam and follower makes it a higher pair.

Chapter 9: Bearings

Bearings – Basic Concept

A **bearing** is a machine element that supports a rotating shaft and permits relative motion while carrying the load, with minimum friction. Bearings are broadly classified as **sliding contact** (journal) bearings and **rolling contact** (anti-friction) bearings such as ball and roller bearings.

Understand in Detail

9.1 Bearings — the 'support' of a rotating shaft

Why does a bicycle wheel turn so easily? Its hub has small steel balls — a **ball bearing**. A **bearing** is a machine element that **supports** a rotating or sliding part (shaft), guides its motion and transfers the load to the frame — keeping **friction and wear to a minimum**.

By direction of load: radial (journal) — load perpendicular to the shaft; **thrust** — load along the shaft axis — **footstep/pivot** (end of a vertical shaft) and **collar** (horizontal shaft, ship propeller). **By contact: sliding (plain/journal)** — surface contact with an oil film between; **rolling (anti-friction)** — balls or rollers, point/line contact.

9.2 Sliding bearings and the secret of lubrication

In a **journal bearing** the part of the shaft that rotates inside the bearing is called the **journal**, and the shell surrounding it the **bush/brass** (usually bronze or **Babbitt metal** — soft, so that the bush wears and not the shaft). Simplest — **solid bearing**; the two-part **split/plummer block** (allows adjustment when worn).

Hydrodynamic lubrication — how the shaft 'floats': at rest the shaft sits on the bottom of the bush. As it starts rotating it drags oil with it, and the oil is squeezed into the **converging wedge-shaped** gap between shaft and bush — the pressure rises so much that the shaft lifts and **floats** on the oil film; metal-to-metal contact ends. This is **thick-film** lubrication — very low friction. At low speed the film thins and becomes **boundary** lubrication — more friction and wear.

Petroff's law (light load, concentric shaft): coefficient of friction $\mu = 2\pi^2 (ZN/p)(r/c)$, where Z = viscosity, N = speed, p = pressure, c = radial clearance. **Bearing characteristic number ZN/p** — the graph of μ vs ZN/p (**Stribeck curve**) shows a minimum point; to its right lies stable thick-film lubrication. **Sommerfeld number** $S = (r/c)^2 (ZN/p)$ — the dimensionless design parameter.

Derivation: Friction torque in thrust bearings

1. **Flat pivot** (radius R, load W): uniform pressure T = (2/3)μWR ; uniform wear T = (1/2)μWR
2. **Conical pivot** (semi-angle α): uniform pressure T = (2/3)μWR/sin α ; uniform wear T = (1/2)μWR/sin α
3. **Collar bearing** (R_1, R_2): formulae like a clutch plate — uniform wear T = μW(R_1 + R_2)/2
4. **Power lost = 2πNT/60**

9.3 Rolling bearings — balls and rollers

In a rolling bearing balls or rollers roll between two **races** (inner — on the shaft, outer — in the housing), kept equally spaced by a **cage (separator)**. Rolling instead of sliding gives **very low starting friction** — so they are called **anti-friction bearings**.

Types: deep groove ball — the most common, radial + some axial load; **self-aligning ball** — tolerates shaft misalignment; **angular contact** — radial + larger axial; **cylindrical roller** — heavy radial load (line contact), no axial; **taper roller** — heavy radial + axial (car wheel hubs); **needle roller** — very thin long rollers, little radial space; **spherical roller** — heavy load + misalignment; **thrust ball** — axial only.

Life: during rotation the races and balls are stressed repeatedly; eventually the surface flakes off by **fatigue** (pitting/spalling). **Rated life L_{10}** — the life (million revolutions) that 90% of bearings will complete. **$L = (C/P)^k$** million revolutions, where C = **basic dynamic load rating** (load for L = 1 million), P = equivalent load, **k = 3 (ball), 10/3 (roller)**. In hours: $L_h = L \times 10^6/(60N)$. Meaning: load doubles → ball bearing life becomes **1/8**!

Example: A ball bearing has dynamic load rating 20 kN, equivalent load 4 kN, speed 1000 rpm. Life in hours?

Solution:

1. $L = (20/4)^3 = 125$ million revolutions
2. $L_h = 125 \times 10^6/(60 \times 1000)$ = **2083 hours**

Note: Sliding vs rolling: sliding — quiet, withstands shocks, unlimited life (if thick-film), low cost, but higher starting friction and continuous lubrication; rolling — low starting friction, little lubrication, standard easily replaceable sizes, but noisy, limited (fatigue) life, sensitive to shocks.

1. Key Points

- **By load direction:** radial (journal) bearings carry load perpendicular to the shaft axis; thrust bearings carry load along the axis. Foot-step (pivot) bearings support the axial load of a vertical shaft.
- **Sliding contact bearings:** surfaces slide on each other with a lubricant film; quiet, good for shock loads, cheap.

- **Rolling contact (anti-friction) bearings:** ball and roller bearings; very low starting friction, easy to replace, life limited by fatigue.
- **Taper roller bearing:** carries both radial and axial loads (used in automobile wheels).
- **Hydrodynamic lubrication:** a thick oil film is created by the **rotation of the shaft** itself, which drags oil into a converging wedge. **Hydrostatic** lubrication uses an external pump.
- **Bearing characteristic number:** μN/p (Z N/p); Sommerfeld number S = $(r/c)^2$ μN/p.
- **Rating life L_{10}:** the life in million revolutions that 90% of a group of identical bearings will complete or exceed. $L_{10} = (C/P)^k$, k = 3 for ball bearings and 10/3 for roller bearings.
- **Bearing designation:** in a bearing number like 6205, the last two digits × 5 give the bore in mm (for 04 and above): 05 × 5 = 25 mm. 00 = 10, 01 = 12, 02 = 15, 03 = 17 mm.
- **Bearing materials:** Babbitt metal (white metal – tin/lead based), bronze, cast iron, nylon, PTFE. Babbitt is the most common lining material.
- **Collar and pivot bearings:** friction torque under uniform wear is less than under uniform pressure, similar to clutches.



Fig 9.1: Ball bearing and types of rolling bearings

2. Important Formulas & Facts

Quantity / Topic	Formula / Fact
Rating life	**$L_{10} = (C/P)^k \times 10^6$ revolutions**
k (ball / roller)	**3 / (10/3)**
Bearing characteristic number	**μN / p**
Sommerfeld number	**S = $(r/c)^2$ μN/p**
Flat pivot – uniform pressure	**T = ⅔ μ W R**
Flat pivot – uniform wear	**T = ½ μ W R**

3. Detailed Explanation

- **Life in hours:** $L_h = L_{10} \times 10^6/(60 \times N)$. Since $L \propto (1/P)^3$ for ball bearings, doubling the load reduces the life to 1/8.
- **Stribeck (μ vs ZN/p) curve:** at low values there is boundary lubrication with high friction; beyond the minimum point thick-film (hydrodynamic) lubrication gives stable operation.
- **Selection:** rolling bearings for intermittent loads, high starting loads and where low maintenance is needed; sliding bearings for heavy shock loads, very high speeds and where quiet running is needed.
- **Babbitt metal:** soft, conformable and embeddable; it allows small dirt particles to embed without scoring the shaft, so it is used as a liner on a stronger shell.
- **Lubricants:** oils for high speed, greases for rolling bearings at moderate speed and where sealing is required.

4. Comparison Table — Sliding vs Rolling Bearings

Basis	Sliding (Journal)	Rolling (Ball/Roller)
Starting friction	High	Very low
Shock load	Withstands well	Less able
Noise	Low	More
Life	Practically unlimited with proper lubrication	Limited by fatigue (L_{10})
Radial space	Less	More
Cost / maintenance	Cheap; needs lubrication care	Costly; little maintenance

5. Solved Numerical Problems

Example 1. A ball bearing has dynamic load rating C = 20 kN and equivalent load P = 5 kN. Find the L_{10} life in million revolutions and in hours at 1000 rpm.

Solution:

1. $L_{10} = (C/P)^3 = (20/5)^3 =$ **64 million revolutions**
2. $L_h = 64 \times 10^6/(60 \times 1000) =$ **about 1067 hours**

Example 2. If the above bearing were a roller bearing, what would be its L_{10} life?

Solution:

1. $L_{10} = (C/P)^{10/3} = 4^{3.33} =$ **about 101.6 million revolutions**

Example 3. Find the bore diameters of bearings 6310 and 6302.

Solution:

1. 6310: last two digits 10, so 10 × 5 = **50 mm**
2. 6302: standard value of 02 = **15 mm** (00 = 10, 01 = 12, 02 = 15, 03 = 17)

Example 4. A flat pivot bearing carries W = 10 kN, μ = 0.05, shaft radius R = 0.1 m at 300 rpm. Find the friction torque and power lost under uniform wear.

Solution:

1. T = ½μWR = 0.5 × 0.05 × 10000 × 0.1 = **25 N·m**
2. Power lost = Tω = 25 × 31.42 = **about 785 W**

6. Frequently Asked Facts

- $L_{10} = (C/P)^3$ for ball bearings and $(C/P)^{10/3}$ for roller bearings.
- Bore of 62xx/63xx bearings = last two digits × 5 mm (from 04 onward).
- Babbitt metal is the common bearing liner material.

7. Note (Confusing Facts)

- L_{10} is the life reached by **90%** of bearings (10% may fail earlier).
- Hydrodynamic film needs shaft rotation; hydrostatic needs an external pump.
- Bore codes 00–03 do not follow the ×5 rule (10, 12, 15, 17 mm).

Remember: One line: taper roller bearing – radial + axial load together. Assertion–Reason: (A) Rolling bearings are called anti-friction bearings. (R) Their starting friction is very low. — Both true; R explains A.

30 Most Important One-Liner Facts — Bearings

1. Function of a bearing?

— To support a shaft and allow motion with minimum friction.

2. Anti-friction bearings are?

— Ball and roller bearings.

3. Bearing for axial load of a vertical shaft?

— Foot-step (pivot) bearing.

4. Bearing for both radial and axial loads?

— Taper roller bearing.

5. Bearing better for shock loads?

— Sliding contact bearing.

6. Bearing with lowest starting friction?

— Rolling contact bearing.

7. L_{10} life definition?

— Life completed by 90% of bearings.

8. Life formula of ball bearing?

— $(C/P)^3$ million revolutions.

9. Value of k for roller bearings?

— 10/3.

10. Effect of doubling load on ball bearing life?

— Becomes 1/8.

11. Life in hours?

— $L_{10} \times 10^6/(60N)$.

12. Bore of bearing 6205?

— 25 mm.

13. Bore code 00?

— 10 mm.

14. Bore code 01?

— 12 mm.

15. Bore code 02?

— 15 mm.

16. Bore code 03?

— 17 mm.

17. Bearing characteristic number?

— μN/p.

18. Sommerfeld number?

— $(r/c)^2$ μN/p.

19. Hydrodynamic lubrication requires?

— Rotation of the shaft.

20. Hydrostatic lubrication requires?

— External pressurised oil supply.

21. Common bearing liner material?

— Babbitt (white) metal.

22. Main constituents of Babbitt?

— Tin or lead with antimony and copper.

23. Why Babbitt is used?

— Soft, conformable, embeddable.

24. Friction torque of flat pivot (uniform wear)?

— ½μWR.

25. Friction torque of flat pivot (uniform pressure)?

— ⅔μWR.

26. A crankshaft in a journal bearing forms?

— Turning pair.

27. Radial space needed by sliding bearings?

— Less.

28. Life of a rolling bearing is limited by?

— Fatigue.

29. Lubricant generally used in rolling bearings?

— Grease.

30. Bearing used in automobile wheel hubs?

— Taper roller bearing.

Practice MCQs — Bearings

Q1. Anti-friction bearings are:

(a) Journal
(b) Ball and roller
(c) Bush
(d) Pivot

✔ **Answer: (b) Ball and roller**

Explanation: Ball and roller bearings have rolling contact and very low friction – anti-friction bearings.

Q2. Life of a ball bearing L =

(a) $(C/P)^2$
(b) $(C/P)^3$
(c) $(C/P)^{10/3}$
(d) C/P

✔ **Answer: (b) $(C/P)^3$**

Explanation: For ball bearings, L = $(C/P)^3$ million revolutions.

Q3. For roller bearings, k =

(a) 3
(b) 10/3
(c) 2
(d) 4

✔ **Answer: (b) 10/3**

Explanation: For roller bearings, the life exponent k = 10/3.

Q4. Bore of bearing 6205:

(a) 20 mm
(b) 25 mm
(c) 52 mm
(d) 5 mm

✔ **Answer: (b) 25 mm**

Explanation: Bore = 05 × 5 = 25 mm.

Q5. Bearing carrying both radial and axial loads:

(a) Needle
(b) Taper roller
(c) Thrust ball
(d) Cylindrical roller

✔ **Answer: (b) Taper roller**

Explanation: The taper roller's conical rollers take radial and thrust loads together.

Q6. Bearing for axial load of a vertical shaft:

(a) Journal
(b) Foot-step (pivot)
(c) Plummer block
(d) Needle

✔ **Answer: (b) Foot-step (pivot)**

Explanation: A foot-step (pivot) bearing supports the axial load at the end of a vertical shaft.

Q7. Hydrodynamic lubrication requires:

(a) External pump
(b) Rotation of shaft
(c) Grease
(d) High load

✔ **Answer: (b) Rotation of shaft**

Explanation: The rotating shaft drags oil into the wedge and builds a pressure film.

Q8. L_{10} life is the life completed by:

(a) 50% bearings
(b) 90% bearings
(c) 10% bearings
(d) 100% bearings

✔ **Answer: (b) 90% bearings**

Explanation: L_{10}: 90% of identical bearings reach or exceed this life.

Q9. Babbitt metal is mainly a:

(a) Bearing material
(b) Gear material
(c) Spring steel
(d) Cutting tool

✔ **Answer: (a) Bearing material**

Explanation: Babbitt (white metal) is a popular bearing material.

Q10. If load is doubled, ball bearing life becomes:

(a) 1/2
(b) 1/4
(c) 1/8
(d) 1/16

✔ **Answer: (c) 1/8**

Explanation: $L \propto (1/P)^3$, so doubling the load gives $(1/2)^3 = 1/8$ life.

SSC JE (Mechanical) Previous Year Questions — Bearings

Q1. Material used for a bearing liner:

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(a) Babbitt metal
(b) Cast steel
(c) High carbon steel
(d) Aluminium

✔ **Answer: (a) Babbitt metal**

Explanation: Babbitt metal is the common liner because it is soft, conformable and embeddable.

Q2. A crankshaft turning in a journal bearing forms a:

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(a) Sliding pair
(b) Turning pair
(c) Rolling pair
(d) Helical pair

✔ **Answer: (b) Turning pair**

Explanation: The shaft rotating in the journal forms a turning pair.

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UPSSSC JE 2016 – Previous Year Questions

Answers as per the answer key published by the Commission

Q1. Velocity ratio of a wheel and differential axle is:

(a) $(d_1d_2)/2D$ (b) $(d_1d_2)/4D$

(c) $2D/(d_1 - d_2)$ (d) $3D/(d_1 - d_2)$

✔ **Answer: (c) $2D/(d_1 - d_2)$**

Explanation: VR of differential wheel and axle = $2D/(d_1 - d_2)$.

Q2. Max and min speeds of a flywheel during a cycle are N_1 and N_2 rpm. The coefficient of steadiness is:

(a) $(N_1 - N_2)/2(N_1 + N_2)$ (b) $(N_1 + N_2)/2(N_1 - N_2)$

(c) $2(N_1 + N_2)/(N_1 - N_2)$ (d) $(N_1 + N_2)/(N_1 - N_2)$

✔ **Answer: (b) $(N_1 + N_2)/2(N_1 - N_2)$**

Explanation: Coefficient of steadiness = $1/C_S = N/(N_1 - N_2) = (N_1 + N_2)/2(N_1 - N_2)$.

Q3. Coefficient of fluctuation of energy of an engine is:

(a) Variation of energy above or below mean resisting torque

(b) Ratio of maximum and minimum energy

(c) Ratio of maximum fluctuation of energy to work done per cycle

(d) None of above

✔ **Answer: (c) Ratio of maximum fluctuation of energy to work done per cycle**

Explanation: C_E = maximum fluctuation of energy ÷ work done per cycle.

Q4. Power transmitted by a belt is maximum when maximum tension is ____ the centrifugal tension.

(a) 2 times (b) 3 times

(c) 4 times (d) 2.5 times

✔ **Answer: (b) 3 times**

Explanation: For maximum power $T_{max} = 3T_c$.

Q5. Maximum permissible velocity of a belt is:

(a) $\sqrt{(Tg/3w)}$ (b) $\sqrt{(2Tg/3w)}$

(c) $\sqrt{(3Tg/2w)}$ (d) $\sqrt{(3Tg/w)}$

✔ **Answer: (a) $\sqrt{(Tg/3w)}$**

Explanation: From $T = 3T_c = 3wv^2/g$, $v = \sqrt{(Tg/3w)}$.

Q6. Initial tension in a stationary belt is:

(a) T_1 (b) T_2

(c) $T_1 + T_2$ (d) $(T_1 + T_2)/2$

✔ **Answer: (d) $(T_1 + T_2)/2$**

Explanation: Initial tension = average of tight and slack side tensions.

Q7. Gears are termed medium velocity gears if their peripheral velocity is:

(a) 1–3 m/s (b) 3–15 m/s

(c) 15–30 m/s (d) 30–50 m/s

✔ **Answer: (b) 3–15 m/s**

Explanation: Pitch-line velocity 3–15 m/s – medium velocity gears.

Q8. If a gear train of 12 gears is used, the driver and follower move in:

(a) Same direction (b) Opposite direction

(c) Both (1) and (2) (d) None of these

✔ **Answer: (b) Opposite direction**

Explanation: With an even number of gears the first and last rotate in opposite directions.

Q9. To transmit power between shafts neither parallel nor intersecting, use:

(a) Spur gear (b) Spiral gear

(c) Crown gear (d) Worm gear

✔ **Answer: (d) Worm gear**

Explanation: Worm gears connect non-parallel, non-intersecting shafts.

Q10. In an involute gear, the normal to the involute is tangent to the:

(a) Pitch circle (b) Base circle

(c) Addendum circle (d) Dedendum circle

✔ **Answer: (b) Base circle**

Explanation: The normal to an involute is always tangent to its base circle.

Q11. If gear and pinion are of the same material, load capacity is decided by:

(a) Gear (b) Pinion

(c) Any one of two (d) None of these

✔ **Answer: (b) Pinion**

Explanation: The pinion has fewer teeth and is weaker, so it decides the load capacity.

Q12. Height of a Watt governor (in metre) is:

(a) $8.95/N^2$ (b) $89.5/N^2$

(c) $895/N^2$ (d) $8.95/N$

✔ **Answer: (c) $895/N^2$**

Explanation: $h = g/\omega^2 = 895/N^2$ m.

Q13. A governor is unstable when the radius of balls ____ as equilibrium speed increases.

(a) Increases
(b) Decreases
(c) Remains constant
(d) Fluctuates

✔ **Answer: (b) Decreases**

Explanation: If the radius decreases as speed increases, the governor is unstable.

Q14. In a Hartnell governor, if spring stiffness is increased, the governor becomes:

(a) More sensitive
(b) Less sensitive
(c) Hunting
(d) Insensitive

✔ **Answer: (b) Less sensitive**

Explanation: Greater stiffness needs a larger speed change to move the sleeve – less sensitive.

Q15. In high-speed engines, the cam follower should move with:

(a) S.H.M.
(b) Uniform velocity
(c) Cycloidal motion
(d) None of the above

✔ **Answer: (c) Cycloidal motion**

Explanation: Cycloidal motion has zero acceleration at the ends – best at high speed.

Q16. Throw of a cam is the maximum distance of the follower from the:

(a) Base circle
(b) Pitch circle
(c) Prime circle
(d) Pitch curve

✔ **Answer: (a) Base circle**

Explanation: Throw is measured from the base circle.

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